Coupling Integrals of Polynomial Cubes at a Welded Elastic Interface

The Static Image in Closed Form, and the Plane Waves for the Rest

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Abstract

In a stratified elastic reference medium the Green's tensor is that of a homogeneous medium plus the waves reflected and transmitted at the interfaces, and it is most naturally available not as a tabulated function but through its plane-wave, up- and down-going, factorisation. A volume-integral method that divides the heterogeneity into cubic cells carrying polynomial fields needs the Galerkin coupling of every pair of cells through this reference. Between cells in planes that do not touch, the plane-wave representation with the cells' moments in closed form is exact, and its integral over the lateral wavenumber converges exponentially. Next to a welded interface it does not: a cell is coupled to itself and to its neighbours through its own mirror image, which touches them, and the reflected coupling converges only algebraically. We show that the singular part is the static field of a point force at the interface, to which the reflected and transmitted kernels tend as $(\omega/\beta q)^2$ at large lateral wavenumber q, and obtain it and its coupling blocks in closed form. Solved in the lateral-wavenumber domain, Rongved's problem gives every entry as $\mathrm{e}^{-q\zeta}/q$ times a polynomial; in space the reflected and transmitted static fields are 356 and 275 terms built on Kelvin's potential and two logarithmic potentials, Mindlin's terms, which at a welded interface are present for every pair of distinct media and which we write as a line of images. The moments of these kernels over a box with the singular point at a vertex reduce, by Euler's identity for homogeneous functions, to faces, edges and edges to infinity, with Hadamard finite parts where the degree is negative and two base integrals where it is zero, $B_1 = \frac{5}{4}\mathrm{Cl}_2(\pi/3) - G$ and $B_2 = \frac{1}{3}\log[(1+\sqrt{2})^2/(2(1+\sqrt{3}))]$, so that every moment is elementary apart from Catalan's constant and one value of Clausen's function. The corner moments and the blocks agree with independent routes to 10^{-15} and 10^{-13} or better, coincide with the blocks of the homogeneous medium to 2 to 8×10^{-15} when the two media are the same, and with the static part subtracted the plane-wave remainder converges as Q^{-4} in the cutoff Q instead of Q^{-2}. The next term of the kernels, of order ω^2 with a coefficient that does not depend on the frequency, is derived in the wavenumber domain; subtracted as well, it leaves a remainder that converges as Q^{-6}, so that a double-precision plane-wave integral truncated at $Qh = 10$ gives the block of a touching cell to 4×10^{-10} at $\omega h/\beta = 1/6$.

1 Introduction

A volume-integral method for elastic waves represents a heterogeneous body as a contrast with a reference medium, divides the contrast into cells, and couples every pair of cells through the Green's tensor of the reference. When the reference is homogeneous the coupling of cubic cells carrying polynomial fields can be obtained in closed form and evaluated to round-off [1]. In seismology the natural reference is stratified, and its Green's tensor is that of a homogeneous medium plus waves reflected and transmitted at the interfaces. The stratified Green's tensor is rarely tabulated: it is computed through its plane-wave representation, in which, at each lateral wavenumber, waves travel up and down through the layers and are reflected and transmitted at each interface [2, 3]. A volume method built on a stratified reference then needs the coupling blocks in a form that uses only this up- and down-going factorisation [4]; tabulating the Green's tensor between all pairs of depths would cost storage quadratic in the number of depth planes, where the plane waves need only linear storage.

For cells in different planes that do not touch, the plane-wave representation with the cells' moments in closed form is exact, and its integral over the lateral wavenumber converges exponentially because the evanescent waves decay between the planes. For cells in touching planes it diverges, and those blocks are the closed forms of the homogeneous medium [1]. At an interface the same failure reaches further. The reflection of a cell's field is the field of its mirror image, and the image of a cell that touches the interface touches the cell itself and its neighbours along the interface. The reflected coupling of every such cell is therefore singular, and its plane-wave integral converges only algebraically.

This paper removes that obstacle. Its contributions are the following.

1. The plane-wave representation of the coupling blocks, stated for the direct, reflected and transmitted couplings with the cells' moments in closed form (Section 2).
2. The identification of the singular part of the reflected and transmitted couplings as the static field of a point force at a welded interface (Rongved's problem), with the dynamic kernels tending to it as $(\omega/\beta q)^2$; subtracting it changes the convergence of the plane-wave remainder from Q^{-2} to Q^{-4} (Sections 2 and 3).
3. The static image and the transmitted static field in space, as 356 and 275 canonical terms on three potentials, with Mindlin's logarithmic terms written as a line of images (Section 3).
4. The coupling blocks of these fields in closed form: the moments over a box with the singular point at a vertex reduce to elementary functions and two constants, B_1 and B_2, which we evaluate in closed form (Sections 4 and 5).
5. The term of order ω^2 of the reflected and transmitted kernels, by perturbation of the static problem, and its subtraction in the wavenumber domain with a frequency-independent coefficient; the remainder then converges as Q^{-6}, and a cutoff low enough for double-precision mode sums suffices (Section 6).
6. Verification of each step against routes that share no step with it (Section 7).

The dynamic remainder, smooth by construction, is left to the plane waves; Section 8 describes how they carry it through a stack of layers.

2 Cells, blocks and the plane-wave representation

Cells and blocks. A cell is a cube of half-width h centred at $\mathbf{x}_c$, with local coordinates $\boldsymbol{\xi} = (\mathbf{x}-\mathbf{x}_c)/h \in [-1,1]^3$. It carries a nine-component state, the displacement and an independent engineering strain, as a polynomial in $\boldsymbol{\xi}$ [5]; the field is tested against products of Legendre polynomials $L_a(\boldsymbol{\xi}) = \prod_i P_{l_{a,i}}(\xi_i)$, and the source density, the contrast times the state, is written in the monomials $m_c(\boldsymbol{\xi}') = \prod_i \xi_i'^{e_{c,i}}$. With $\mathbf{P}(\mathbf{x},\mathbf{x}')$ the 9×9 point propagator of the reference medium, which maps a point source of force and engineering stress at $\mathbf{x}'$ to the displacement and engineering strain at $\mathbf{x}$, a field cell V_m and a source cell V_n couple through the blocks

$$\mathbf{K}_{ac} = \int_{V_m}\int_{V_n} L_a(\boldsymbol{\xi})\,\mathbf{P}(\mathbf{x},\mathbf{x}')\,m_c(\boldsymbol{\xi}')\,\mathrm{d}^3x\,\mathrm{d}^3x'. \tag{1}$$

In a homogeneous medium $\mathbf{P}$ depends on $\mathbf{x}-\mathbf{x}'$ only, and with $\mathbf{x} = \mathbf{x}_m + h\boldsymbol{\xi}$, $\mathbf{x}' = \mathbf{x}_n + h\boldsymbol{\xi}'$ the block becomes an integral over the separation $\mathbf{s} = h(\boldsymbol{\xi}-\boldsymbol{\xi}') \in [-2h,2h]^3$ against the weight $W_{ac}(\mathbf{s}) = \prod_i w_{l_{a,i}e_{c,i}}(s_i)$, with $w_{\alpha\beta}(s) = h\int v^\alpha (v - s/h)^\beta\,\mathrm{d}v$ the cross-correlation of the two cells' one-dimensional factors, a piecewise polynomial on eight boxes [1]. The kernel is singular at $\mathbf{s} = -(\mathbf{x}_m - \mathbf{x}_n)$, which for touching cells is a vertex of some of the boxes.

The plane-wave representation. Take coordinates in the order (z,x,y), z the depth, and a lateral wavenumber $\mathbf{k} = (k_x,k_y) \in \mathbb{R}^2$, $q = |\mathbf{k}|$. For each wave $\nu \in \{\mathrm{P},\mathrm{SV},\mathrm{SH}\}$ of speed c_ν ($c_\mathrm{P} = \alpha$, $c_\mathrm{SV} = c_\mathrm{SH} = \beta$) let

$$k_{z,\nu} = \left(\omega^2/c_\nu^2 - q^2\right)^{1/2}, \quad \operatorname{Im} k_{z,\nu} \geq 0, \qquad \mathbf{k}_\nu^\sigma = (\sigma k_{z,\nu}, k_x, k_y), \quad \sigma = \pm 1,$$

the wave vector travelling down ($\sigma = +1$) or up ($\sigma = -1$). Its polarisations are

$$\mathbf{e}_\mathrm{P}^\sigma = \frac{\alpha}{\omega}\,\mathbf{k}_\mathrm{P}^\sigma, \qquad \mathbf{e}_\mathrm{SH} = \frac{(0,-k_y,k_x)}{q}, \qquad \mathbf{e}_\mathrm{SV}^\sigma = \frac{\beta}{\omega}\,\mathbf{k}_\mathrm{SV}^\sigma \times \mathbf{e}_\mathrm{SH},$$

with $\mathbf{e}_\mathrm{SH} = (0,1,0)$ at $\mathbf{k} = \mathbf{0}$. Each is a unit vector in the bilinear sense, $\mathbf{e}\cdot\mathbf{e} = 1$, which, unlike $\mathbf{e}\cdot\bar{\mathbf{e}} = 1$, continues analytically into the evanescent range $q > \omega/c_\nu$. The state of the wave is the nine-vector $\mathbf{d}_\nu^\sigma = (\mathbf{e},\boldsymbol{\varepsilon})$ of its displacement and engineering strain, $\varepsilon_{ij} = \frac{\mathrm{i}}{2}(k_i e_j + k_j e_i)$ with $\mathbf{k} = \mathbf{k}_\nu^\sigma$ and the three shears doubled, and $\mathbf{M} = \operatorname{diag}(1,1,1,1,1,1,\frac{1}{2},\frac{1}{2},\frac{1}{2})$ pairs the engineering-stress entries of a source with the engineering-strain entries of a state. In a homogeneous medium of density ρ, for $\sigma(z-z') > 0$,

$$\mathbf{P}(\mathbf{x},\mathbf{x}') = \frac{1}{4\pi^2}\int_{\mathbb{R}^2}\sum_\nu w_\nu\,\mathbf{d}_\nu^\sigma\,\mathbf{d}_\nu^{\sigma\mathsf{T}}\mathbf{M}\,\mathrm{e}^{\mathrm{i}\mathbf{k}_\nu^\sigma\cdot(\mathbf{x}-\mathbf{x}')}\,\mathrm{d}^2k, \qquad w_\nu = \frac{\mathrm{i}}{2\rho c_\nu^2 k_{z,\nu}}, \tag{2}$$

the source entering through $\mathbf{d}_\nu^{\sigma\mathsf{T}}\mathbf{M}$ and the field through $\mathbf{d}_\nu^{\sigma}$, as reciprocity requires. For cells in planes that do not touch, $|z_m - z_n| \geq 4h$, set $\sigma = \operatorname{sgn}(z_m - z_n)$; Eq. (1) then becomes

$$\mathbf{K}_{ac} = \frac{1}{4\pi^2} \int_{\mathbb{R}^2} \sum_\nu w_\nu \, \Lambda_a(\mathbf{k}_\nu^\sigma) \, \mu_c(-\mathbf{k}_\nu^\sigma) \, \mathbf{d}_\nu^\sigma \, \mathbf{d}_\nu^{\sigma\mathsf{T}} \mathbf{M} \, \mathrm{e}^{\mathrm{i}\mathbf{k}_\nu^\sigma \cdot (\mathbf{x}_m - \mathbf{x}_n)} \, \mathrm{d}^2 k, \tag{3}$$

with the scalar moments of the two cells, both in closed form for complex arguments [1],

$$\Lambda_a(\boldsymbol{\kappa}) = h^3 \int_{[-1,1]^3} L_a(\boldsymbol{\xi}) \, \mathrm{e}^{\mathrm{i}h\boldsymbol{\kappa}\cdot\boldsymbol{\xi}} \, \mathrm{d}^3\xi = h^3 \prod_i 2\,\mathrm{i}^{l_{a,i}} j_{l_{a,i}}(h\kappa_i), \tag{4}$$

$$\mu_c(\boldsymbol{\kappa}) = h^3 \int_{[-1,1]^3} m_c(\boldsymbol{\xi}') \, \mathrm{e}^{\mathrm{i}h\boldsymbol{\kappa}\cdot\boldsymbol{\xi}'} \, \mathrm{d}^3\xi' = h^3 \prod_i F_{e_{c,i}}(h\kappa_i), \qquad F_n(t) = \int_{-1}^{1} \xi^n \mathrm{e}^{\mathrm{i}t\xi} \, \mathrm{d}\xi, \tag{5}$$

j_l the spherical Bessel functions and F_n a finite combination of them.

Reflection and transmission. Let medium A, with Lamé constants λ_A, μ_A and density ρ_A, fill $z < 0$ and medium B fill $z > 0$, welded at $z = 0$ (Figure 1). With the amplitudes referred to the interface and the polarisations of each medium, a down-going wave ν in A gives the up-going wave μ in A with amplitude $R_{\mu\nu}(\mathbf{k})$ and the down-going wave μ in B with amplitude $T_{\mu\nu}(\mathbf{k})$, obtained at each $\mathbf{k}$ from the continuity of displacement and traction, a 6×6 linear system [3]. For $\mathbf{x}$ and $\mathbf{x}'$ both in A the reflected propagator is

$$\mathbf{P}^R(\mathbf{x}, \mathbf{x}') = \frac{1}{4\pi^2} \int_{\mathbb{R}^2} \sum_{\mu,\nu} \mathbf{d}_\mu^{A-} \, R_{\mu\nu} \, w_\nu^A \, \mathbf{d}_\nu^{A+\mathsf{T}} \mathbf{M} \, \mathrm{e}^{\mathrm{i}(\mathbf{k}_\mu^{A-}\cdot\mathbf{x} - \mathbf{k}_\nu^{A+}\cdot\mathbf{x}')} \, \mathrm{d}^2 k, \tag{6}$$

and for $\mathbf{x}'$ in A and $\mathbf{x}$ in B the whole coupling is the transmitted one,

$$\mathbf{P}^T(\mathbf{x}, \mathbf{x}') = \frac{1}{4\pi^2} \int_{\mathbb{R}^2} \sum_{\mu,\nu} \mathbf{d}_\mu^{B+} \, T_{\mu\nu} \, w_\nu^A \, \mathbf{d}_\nu^{A+\mathsf{T}} \mathbf{M} \, \mathrm{e}^{\mathrm{i}(\mathbf{k}_\mu^{B+}\cdot\mathbf{x} - \mathbf{k}_\nu^{A+}\cdot\mathbf{x}')} \, \mathrm{d}^2 k, \tag{7}$$

the superscripts naming the medium whose wave numbers and polarisations are used. The blocks follow as in Eq. (3), with $\Lambda_a(\mathbf{k}_\mu^+)\,\mu_c(-\mathbf{k}_\nu^+)$ in place of $\Lambda_a\mu_c$ and the phase $\mathrm{e}^{\mathrm{i}(\mathbf{k}_\mu\cdot\mathbf{x}_m - \mathbf{k}_\nu\cdot\mathbf{x}_n)}$. The vertical phase of the reflected term is $\mathrm{e}^{\mathrm{i}(k_{z,\mu}|z| + k_{z,\nu}|z'|)}$: its decay on the evanescent branch is set by the distance of the receiver from the source's mirror image $\mathbf{x}'^* = (-z', x', y')$.

Where the plane waves fail. For cells in touching planes, $|z_m - z_n| = 2h$, the integral (3) does not converge: on the evanescent branch $k_{z,\nu} = \mathrm{i}\kappa$, $\kappa \to \infty$, the moments Λ_a and μ_c grow as $\mathrm{e}^{\kappa h}$ each and cancel the decay $\mathrm{e}^{-2\kappa h}$ of the phase. In the reflected coupling the relevant distance is that from the image, $|z_m + z_n|$, which is $2h$ for any two cells that touch the interface, and the same cancellation occurs (Figure 1a). The integrand then decays only algebraically in q, and so does the error of a truncated integral. Table 1 shows it for a cell that touches the interface, coupled to its own image: the contribution of each shell $Q < qh < 2Q$ falls by a factor of four from one shell to the next, so the error of an integral truncated at Q falls as Q^{-2}.

The singular part of the integrand is its behaviour at large q. There $k_{z,\nu} = \mathrm{i}q\,[1 - \omega^2/(2c_\nu^2 q^2) + O(q^{-4})]$, and the reflected and transmitted kernels tend to the static kernel of Section 3. At fixed

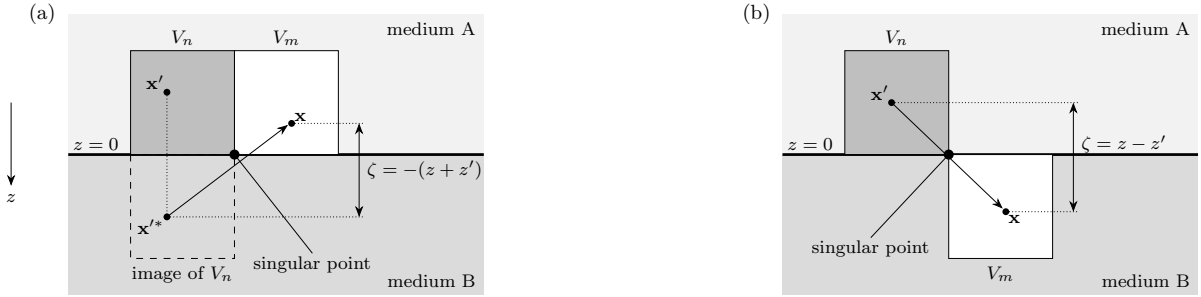


Figure 1. Cells at a welded interface, in a vertical section; z is depth. (a) Two cells in medium A that touch the interface and each other. The reflected coupling is the field of the source cell's mirror image (dashed), at the depth $\zeta = -(z + z')$ below the field point, and the image touches the field cell at the singular point. (b) A source cell in A and a field cell in B that touch at a point of the interface. The transmitted coupling depends on $\zeta = z - z'$, and the singular point is where the cells touch.

depths the difference is of relative order $\omega^2\zeta/(c^2q)$, since the correction to $k_{z,\nu}$ multiplies the depth ζ in the phase; in the block of cells that touch the interface the depths that matter are of order $1/q$, and the difference of the integrands is of relative order $(\omega/\beta q)^2$ (about $2(\omega/\beta q)^2$ for the cell of Table 1, from $qh = 20$ to 1280). Writing $\mathbf{K}^R = \mathbf{K}_0^R + (\mathbf{K}^R - \mathbf{K}_0^R)$, with $\mathbf{K}_0^R$ the block of the static image, the first term is obtained in closed form (Section 4) and the second by the plane-wave integral with the static spectrum subtracted under the integral sign. The remainder converges as Q^{-4} (Table 1): the subtraction gains exactly the two orders of $(\omega/\beta q)^2$, and at $Qh = 10$ it leaves 1.3×10^{-6} of the block where the truncated integral leaves 8×10^{-3}.

The kernels must be evaluated with care at large q. Written as a sum over the modes ν, as in Eqs. (6) and (7), the P and SV terms cancel as $(cq/\omega)^4$, because their bilinear-unit polarisations grow as cq/ω while their sum tends to the finite static kernel. In double precision the reflected kernel is then in error by 6×10^{-7} at $qh = 20$ for the cell of Table 1, and above $qh \approx 100$ the remainder after the subtraction is mostly rounding error. Table 1 was therefore computed with the kernels in 40-digit arithmetic, which agree with 60 digits to 3×10^{-16} at $qh = 640$. A production evaluation in double precision therefore needs a cutoff below $qh \approx 20$; Section 6 removes the next term of the expansion so that such a cutoff is accurate enough.

	reflected kernel		less its static part	
shell qh	contribution	order	contribution	order
10–20	5.8×10^{-3}		1.2×10^{-6}	
20–40	1.4×10^{-3}	2.02	7.7×10^{-8}	4.00
40–80	3.3×10^{-4}	2.13	4.3×10^{-9}	4.17
80–160	8.5×10^{-5}	1.95	2.8×10^{-10}	3.90
160–320	2.1×10^{-5}	2.01	1.8×10^{-11}	4.01
320–640	5.3×10^{-6}	2.00	1.1×10^{-12}	4.00

Table 1. The plane-wave integral of the reflected block of a cell touching the interface, coupled to its own image (field function 1, source monomial 1, media of Section 7, $\omega h/\beta = 0.17$): the contribution of each shell of the lateral wavenumber, relative to the block, and the local order $\log_2$ of the ratio of successive contributions. The kernels are evaluated in 40-digit arithmetic, the azimuthal integral by a trapezoid rule of $4qh + 64$ points and the radial one by Gauss rules on fixed panels; halving the panels and doubling the azimuthal rule changes the shell 160–320 by 6×10^{-20}.

3 The static field of a point force at a welded interface

In the lateral-wavenumber domain. Let a unit point force act in A at depth $z' < 0$. In the frame whose x axis lies along $\mathbf{k}$, the static Navier equations of an isotropic medium have the solutions $(a + bz)\,\mathrm{e}^{\pm qz}$, the double root replacing the separate P and S waves of the dynamic problem. The field in A is Kelvin's field plus a reflected field regular as $z \to -\infty$; the field in B is transmitted and regular as $z \to \infty$; displacement and traction on the planes $z = \text{const}$ are continuous at $z = 0$. This is the problem of Rongved [6], solved here in the lateral-wavenumber domain. Its solution has a simple structure: every entry of the reflected tensor is $\mathrm{e}^{q(z+z')}/q$ times a polynomial of degree at most one in each of qz and qz', and every entry of the transmitted tensor is $\mathrm{e}^{-q(z-z')}/q$ times a polynomial in qz and qz', the coefficients rational in the Lamé constants. When the two media coincide the reflected tensor vanishes and the transmitted one is Kelvin's.

In space. With $\zeta = -(z + z') > 0$ for the reflected field and $\zeta = z - z' > 0$ for the transmitted one, $\boldsymbol{\rho}$ the lateral separation and $R = (|\boldsymbol{\rho}|^2 + \zeta^2)^{1/2}$, the transform to space follows from

$$\frac{1}{4\pi^2}\int_{\mathbb{R}^2} \mathrm{e}^{\mathrm{i}\mathbf{k}\cdot\boldsymbol{\rho}}\, q^m \mathrm{e}^{-q\zeta}\,\mathrm{d}^2k = \begin{cases} (-\partial_\zeta)^{m+1}\Phi_0, & m \geq -1, \\ \Phi_1, & m = -2, \\ \Phi_2, & m = -3, \end{cases} \tag{8}$$

with the three potentials

$$\Phi_0 = \frac{1}{2\pi R}, \qquad \Phi_1 = -\frac{\log(R+\zeta)}{2\pi}, \qquad \Phi_2 = \frac{\zeta\log(R+\zeta) - R}{2\pi}, \tag{9}$$

each the ζ-antiderivative of the one before, $\partial_\zeta\Phi_1 = -\Phi_0$ and $\partial_\zeta\Phi_2 = -\Phi_1$. Rotating from the frame of $\mathbf{k}$ to fixed axes brings factors k_h/q; since $\mathrm{i}k_h$ becomes ∂_{ρ_h}, an entry $(k_h/q)f(q)$ is ∂_{ρ_h} of the transform of $f/(\mathrm{i}q)$, and an entry $(k_hk_l/q^2)f(q)$ is $-\partial_{\rho_h}\partial_{\rho_l}$ of the transform of f/q^2. These divisions by q are what produce the powers $m = -2$ and -3, and so the logarithmic potentials. They are the terms that Mindlin [7] found for the half-space with a free surface; at a welded interface they are present for every pair of media, with coefficients that vanish only when the media coincide.

The potentials as finite parts. The ladder continues below $m = -3$, and the next order in the frequency (Section 6) needs it. For $m > -2$ the transform is the Laplace transform of a Bessel function [8],

$$\frac{1}{2\pi}\int_0^\infty q^{m+1}\mathrm{e}^{-q\zeta}J_0(q\rho)\,\mathrm{d}q = \frac{\Gamma(m+2)}{2\pi}\,R^{-m-2}\,P_{m+1}\!\left(\frac{\zeta}{R}\right), \tag{10}$$

P_ν the Legendre function of the first kind, and at $m = -1$ it is Φ_0. At $m = -1 - j$, $j \geq 1$, the integral diverges at $q = 0$ and Γ has a pole. We define Φ_j as Hadamard's finite part, the coefficient of ε^0 in the Laurent series of Eq. (10) at $m = -1 - j + \varepsilon$: the analytic continuation in the power of q, the same convention by which Euler's identity continues the corner moments in the degree

(Section 4). With $n = j-1$, $P_{-j+\varepsilon} = P_{n-\varepsilon}$ and the derivative of P_ν in its degree at an integer,

$$\frac{\partial P_\nu(x)}{\partial \nu}\Big|_{\nu=n} = P_n(x)\Big[\log\frac{1+x}{2} + 2\psi(2n+1) - 2\psi(n+1)\Big] + 2\sum_{k=0}^{n-1}(-1)^{n+k}\frac{2k+1}{(n-k)(n+k+1)}\,P_k(x), \quad (11)$$

the finite part is

$$\Phi_j = \frac{(-1)^n}{2\pi\, n!}\Big\{\Big[c_j - \log\frac{R+\zeta}{2}\Big]R^n P_n\Big(\frac{\zeta}{R}\Big) - 2\sum_{k=0}^{n-1}(-1)^{n+k}\frac{2k+1}{(n-k)(n+k+1)}\,R^n P_k\Big(\frac{\zeta}{R}\Big)\Big\}, \qquad c_j = 3\psi(j) - 2\psi(2j-1), \quad (12)$$

ψ the digamma function. Each $R^n P_k(\zeta/R)$ is a polynomial in R and ζ, so every potential is a polynomial times $\log(R+\zeta)$ plus a polynomial; for instance

$$\begin{aligned} 2\pi\,\Phi_3 &= \Big(\tfrac{1}{4}R^2 - \tfrac{3}{4}\zeta^2\Big)\Big(\log\tfrac{R+\zeta}{2} + \gamma\Big) - \tfrac{1}{4}R^2 + \tfrac{3}{4}R\zeta + \tfrac{1}{4}\zeta^2, \\ 2\pi\,\Phi_4 &= \Big(\tfrac{5}{12}\zeta^3 - \tfrac{1}{4}R^2\zeta\Big)\Big(\log\tfrac{R+\zeta}{2} + \gamma\Big) + \tfrac{1}{9}R^3 + \tfrac{1}{4}R^2\zeta - \tfrac{5}{12}R\zeta^2 - \tfrac{1}{4}\zeta^3, \end{aligned} \quad (13)$$

γ Euler's constant. The ladder $\partial_\zeta \Phi_{j+1} = -\Phi_j$ holds for the finite parts, since differentiation under the integral commutes with the continuation, and every Φ_j is harmonic. The residue at the pole is $(-1)^n R^n P_n(\zeta/R)/(2\pi\, n!)$, a solid harmonic: it is exactly the freedom that a cutoff at small q would leave in the transform, and the finite part fixes its coefficient. The potentials of Eq. (9) differ from these only by such harmonics: the finite parts are $\Phi_1 + (\log 2 - \gamma)/2\pi$ and $\Phi_2 - \zeta(\log 2 - \gamma)/2\pi$. Since Φ_1 and Φ_2 occur in the static field only under at least one and two lateral derivatives respectively, which annihilate the difference, the static blocks are the same with either. A term $\partial^\alpha \Phi_j$ needs the finite part at all only when its lateral derivatives are fewer than j: its integrand then behaves as $q^{\alpha_x+\alpha_y-1-j}$ at small q and is not integrable in the plane.

Canonical terms. The point propagator differentiates the Green's tensor: a receiver derivative ∂_z acts on the explicit z and on ζ, as $\partial/\partial z - \partial_\zeta$ for the image and $\partial/\partial z + \partial_\zeta$ for the transmitted field, and similarly for source derivatives and the lateral ones. Collected, the 9×9 static image is a sum of 356 terms

$$c(\lambda_A, \mu_A, \lambda_B, \mu_B)\; z^A z'^B\; \partial_{\rho_x}^{\alpha_x}\partial_{\rho_y}^{\alpha_y}\partial_\zeta^{\alpha_z}\Phi_j(\boldsymbol{\rho}, \zeta), \quad (14)$$

in 61 families $(j, \boldsymbol{\alpha})$: Φ_0 with up to four derivatives, and Φ_1 and Φ_2 under lateral derivatives only (one to four, and two to four), since a ζ-derivative of a logarithmic potential is the potential below it. The transmitted field has 275 terms in 46 families of the same kind. Every kernel $\partial^\alpha \Phi_j$ is homogeneous in $(\boldsymbol{\rho}, \zeta)$, of degree $-1-|\boldsymbol{\alpha}|$, $-|\boldsymbol{\alpha}|$ and $1-|\boldsymbol{\alpha}|$ for $j = 0, 1, 2$.

A line of images. Every logarithmic term carries a lateral derivative and therefore vanishes as $\zeta \to \infty$. Integrating the relations between the potentials from ζ to infinity gives

$$\partial^\alpha \Phi_1(\boldsymbol{\rho}, \zeta) = \int_\zeta^\infty \partial^\alpha \Phi_0(\boldsymbol{\rho}, t)\,\mathrm{d}t, \qquad \partial^\alpha \Phi_2(\boldsymbol{\rho}, \zeta) = \int_\zeta^\infty (t-\zeta)\,\partial^\alpha \Phi_0(\boldsymbol{\rho}, t)\,\mathrm{d}t. \quad (15)$$

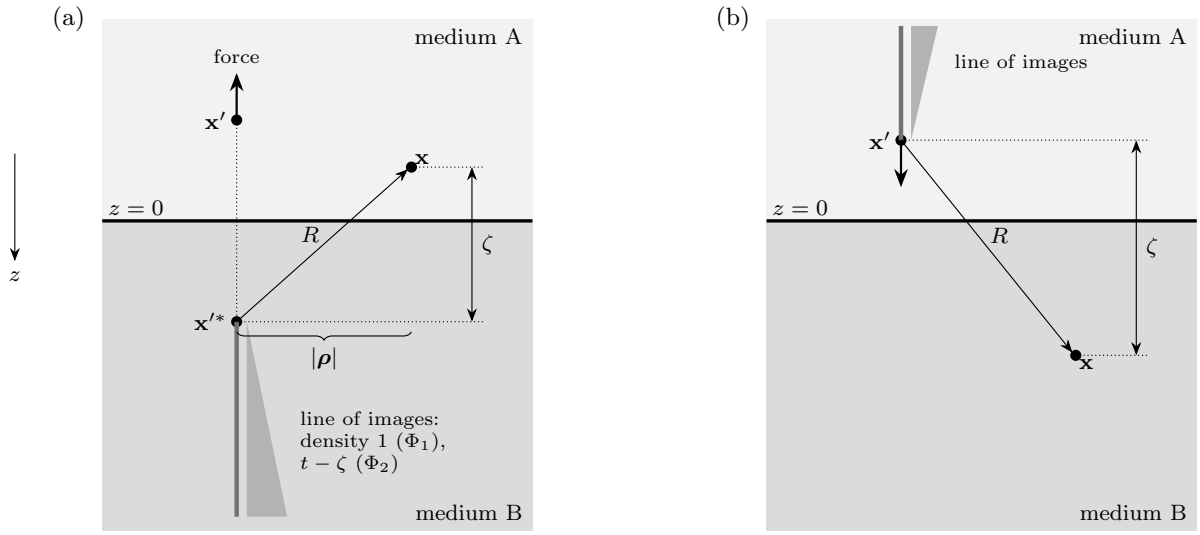


Figure 2. The image sources of the static fields at a welded interface, in a vertical section; z is depth. (a) Reflection: a point force at $\mathbf{x}'$ in medium A is seen from a receiver $\mathbf{x}$ in A through its Kelvin image at the mirror point $\mathbf{x}'^*$, at the distance R and the depth $\zeta = -(z + z')$ below the receiver, and through Mindlin's terms, a line of images on the vertical half-line beyond $\mathbf{x}'^*$ with uniform density (Φ_1) and density growing as $t - \zeta$ (Φ_2), Eq. (15). (b) Transmission: the field in B has no image point; its potentials are centred on the source itself, with $\zeta = z - z'$, and its line of images lies on the vertical half-line above $\mathbf{x}'$.

Mindlin's terms are thus Kelvin's image spread along the vertical half-line beyond the image point, with uniform and with linearly growing density (Figure 2). This is the form in which their moments are computed.

4 Coupling blocks with the static image

The separation form. For two cells in A the image kernel depends on the lateral differences $\rho_h = x_h - x'_h$ but on the vertical sum $\zeta = -(z + z')$. The block is therefore an integral over (ρ_x, ρ_y, ζ) against a weight that is a cross-correlation of the cells' factors in the two lateral coordinates, as in the homogeneous medium, and a convolution in the vertical, $u_{\alpha\beta}(\tau) = \int v^\alpha (\tau - v)^\beta \, \mathrm{d}v$ in $\tau = v + v'$, into which the factors z^A and z'^B of Eq. (14) are expanded. The weights are piecewise polynomials with rational coefficients, computed exactly. The domain divides into boxes on which the weight is a polynomial.

The singular point. The image kernel is singular only at $\boldsymbol{\rho} = 0$, $\zeta = 0$, which is reached when both cells touch the interface and touch each other laterally (Figure 1a), and then at a vertex of the boxes that hold it. There the vertical weight vanishes as ζ^{1+A+B}, and for every one of the 356 terms the integrand is at worst $O(R^{-2})$: it is absolutely integrable, with no distribution and no derivative to be moved onto the weight. Boxes away from the singular point have smooth integrands and are integrated by Gauss rules in coordinates centred on each box. On a box with the singular point at a vertex, reflected onto the positive octant, every term requires the corner

moments

$$M_{j,\alpha}[p,q,r] = \int_0^a\int_0^b\int_0^c \rho_x^p\rho_y^q\zeta^r\,\partial^\alpha\Phi_j(\boldsymbol{\rho},\zeta)\,\mathrm{d}\zeta\,\mathrm{d}\rho_y\,\mathrm{d}\rho_x, \tag{16}$$

with $a = b = c = 2h$. They are universal: by homogeneity they scale as a power of h times a number independent of h, the frequency and the media.

Kelvin's potential. For $j = 0$, $\partial^\alpha R^{-1} = P_\alpha(\boldsymbol{\rho},\zeta)\,R^{-2|\alpha|-1}$ with P_α a homogeneous polynomial of degree $|\boldsymbol{\alpha}|$, so the moments are combinations of the master integrals

$$I[p,q,r;n] = \int_0^a\int_0^b\int_0^c x^p y^q z^r\,(x^2+y^2+z^2)^{n/2}\,\mathrm{d}z\,\mathrm{d}y\,\mathrm{d}x, \tag{17}$$

here for odd n down to -9. The integrand is homogeneous of degree $p+q+r+n$ about the singular point, and Euler's identity with the divergence theorem reduces the box to its three far faces, the faces through the singular point carrying no flux; on a face the same step in two dimensions reduces to edges, and on an edge the integral is elementary [1]. One transcendental function enters, through the face integral of R^{-3}: $\int_0^b\int_0^c R^{-3}\,\mathrm{d}z\,\mathrm{d}y = a^{-1}\arctan\big(bc/(a\sqrt{a^2+b^2+c^2})\big)$ on the face $x = a$, which is continued to lower n by differentiation with respect to a.

The logarithmic potentials. By Eq. (15) the ζ-integral of the box can be done first, which leaves Kelvin's potential against a polynomial weight in t over the box and over the semi-infinite column $[0,a]\times[0,b]\times[c,\infty)$ above it. On $[0,c]$ the weight is $t^{r+1}/(r+1)$ ($j=1$) or $t^{r+2}/((r+1)(r+2))$ ($j=2$); on the column it is $c^{r+1}/(r+1)$, or $t\,c^{r+1}/(r+1) - c^{r+2}/(r+2)$. The column is the semi-infinite box less the finite one. Euler's identity holds on the semi-infinite box, the face at infinity carrying no flux, and its faces reduce to edges that run to infinity,

$$\int_0^\infty z^r(A^2+z^2)^{n/2}\,\mathrm{d}z = \tfrac{1}{2}\,A^{r+n+1}\,\mathrm{B}\Big(\tfrac{r+1}{2},-\tfrac{r+n+1}{2}\Big), \tag{18}$$

Beta functions at half-integers, elementary. Two cases need care. Where the degree $p+q+r+n+3$ of a column term is negative, each of the two boxes holds an integral that diverges at the singular corner; Euler's identity then gives Hadamard's finite part of each, and the two finite parts differ by the exact integral over the column, because their singular contributions at the shared corner are equal. Where the degree is zero the identity is degenerate, and the column is reduced instead by integration by parts in t, then in ρ_x and ρ_y, whose boundary terms are faces at height c and strips at $\rho_x = a$ or $\rho_y = b$, away from the singular point. Two base integrals remain:

$$B_1 = \int_0^a\int_0^b\int_c^\infty R^{-3}\,\mathrm{d}t\,\mathrm{d}\rho_y\,\mathrm{d}\rho_x, \qquad B_2 = \int_0^a\int_0^b\int_c^\infty \rho_x\rho_y\,R^{-5}\,\mathrm{d}t\,\mathrm{d}\rho_y\,\mathrm{d}\rho_x. \tag{19}$$

The base integrals. Both are of degree zero, so they depend only on the ratios of the sides, which here are equal. The t-integrals are elementary, and in polar coordinates (r,θ) in the $\boldsymbol{\rho}$-plane

so are the radial ones: with $c = 1$ and $s = \sqrt{1 + r^2}$,

$$B_1 = \int \log \frac{1+s}{2}\, \mathrm{d}\theta, \qquad B_2 = \int \cos\theta \sin\theta \Big[\frac{1}{3s} + \frac{2}{3}\log(1+s) - \frac{1}{3} - \frac{2}{3}\log 2\Big]\mathrm{d}\theta,$$

with $r = \sec\theta$ on $[0, \pi/4]$ and $r = \csc\theta$ on $[\pi/4, \pi/2]$, the two halves equal by symmetry. For B_2 the substitution $v = \cos^2\theta$ makes the remaining integral elementary. For B_1 the integral $2\int_0^{\pi/4} \log\big(\frac{1}{2}(1 + \sqrt{1 + \sec^2\theta})\big)\mathrm{d}\theta$ evaluates to Catalan's constant G, $\pi \log 2$, and the imaginary parts of $\mathrm{Li}_2(2^{-1/2}\mathrm{e}^{\mathrm{i}\pi/12})$ and $\mathrm{Li}_2(2^{-1/2}\mathrm{e}^{7\mathrm{i}\pi/12})$. Kummer's formula writes $\operatorname{Im}\mathrm{Li}_2(r\mathrm{e}^{\mathrm{i}\theta})$ in Clausen functions $\mathrm{Cl}_2(\theta) = \operatorname{Im}\mathrm{Li}_2(\mathrm{e}^{\mathrm{i}\theta})$ through the auxiliary angle $\omega = \arctan\big(r\sin\theta/(1 - r\cos\theta)\big)$ [9], and at both points $\omega = \pi/6$. The logarithms then cancel, and with $\mathrm{Cl}_2(\pi/2) = G$ and the duplication formula $\mathrm{Cl}_2(2\theta) = 2\mathrm{Cl}_2(\theta) - 2\mathrm{Cl}_2(\pi - \theta)$ at $\theta = \pi/6$,

$$B_1 = \tfrac{5}{4}\,\mathrm{Cl}_2\big(\tfrac{\pi}{3}\big) - G = 0.35271\,14138\ldots, \qquad B_2 = \tfrac{1}{3}\log\frac{(1+\sqrt{2})^2}{2(1+\sqrt{3})} = 0.02151\,58182\ldots. \tag{20}$$

Every corner moment is therefore elementary apart from the two constants G and $\mathrm{Cl}_2(\pi/3)$. All of them are universal numbers, computed once and stored.

A well-conditioned evaluation. The corner moments are monomial moments about the singular corner, and contracting them with the weights expanded in powers about that corner is ill-conditioned in the way the power moments of a homogeneous medium are [1]: at degree two it loses about a digit (Table 7). The blocks are therefore evaluated in a well-conditioned basis, and the corner moments serve as an independent check. On a box away from the singular point the weights are written, exactly, in Legendre polynomials of the box's own coordinates and contracted with the Legendre moments of the kernel, which a tensor Gauss rule gives. A box anchored at the singular point is split into three Duffy pyramids about it [10]. In pyramid k, with $t_k = \rho$ and $t_j = \rho v_j$ for the other two axes in units of the box's sides, every kernel $\partial^\alpha \Phi_j$ is ρ^d times its value on the far face, d its degree of homogeneity, and each one-dimensional weight is $(A_i\rho w_i)^{m_i} r_i(A_i\rho w_i)$, its zero of order m_i at the corner factored out exactly and r_i written in Legendre polynomials. The radial integrand is then $\rho^{2+d+\sum m_i}$ times a polynomial, integrated exactly, and the face integrand is smooth, the far face lying at a distance of two half-widths from the singular point. Two details decide the last digit. The Gauss weights are taken from a 30-digit table, since those of standard library routines carry relative errors growing with the order, of 4×10^{-15} at 32 points and up to 7×10^{-13} at 40; and the tensor-product sums are carried out one axis at a time, since a single running sum over the 32^3 nodes of a box loses about 2×10^{-15}.

5 Cells on opposite sides of the interface

For a source cell in A and a field cell in B the static coupling is the transmitted field, the whole of it, with no direct part. Its potentials are those of Eq. (9) with $\zeta = z - z' = |z| + |z'|$, and the singular point is where the two cells touch at the interface (Figure 1b). The 275 canonical terms are integrable at that point, as the image's are. In the separation form the vertical weight is now a cross-correlation in $v - v'$, like the lateral ones, and the corner moments are those of Section 4.

The other arrangements follow by symmetry. A field cell in A with a source in B, and two cells in B, are the problems above reflected in the plane $z = 0$ with the two media exchanged. Under the reflection the displacement u_z and the shear strains γ_{zx}, γ_{zy} change sign, so the block is conjugated by $\mathrm{diag}(-1,1,1,1,1,1,1,-1,-1)$ in the order $(u_z, u_x, u_y, \varepsilon_{zz}, \varepsilon_{xx}, \varepsilon_{yy}, \gamma_{xy}, \gamma_{zx}, \gamma_{zy})$, and each cell polynomial is multiplied by (-1) to its power of ξ_z.

6 The next term: the order ω^2

The static image removes the part of the reflected integrand that does not decay. What remains decays as the square of the frequency over the wavenumber, and its leading term can be removed the same way. Let the reflected displacement tensor of Section 3, in the frame of $\mathbf{k}$, be expanded at fixed (q, z, z') in powers of the frequency,

$$\mathbf{G}^R(q; z, z'; \omega) = \mathbf{G}_0^R + \omega^2\, \mathbf{G}_1^R + O(\omega^4), \tag{21}$$

with $\mathbf{G}_0^R$ the static image. With $\mathbf{u} = \mathbf{u}_0 + \omega^2 \mathbf{u}_1 + \dots$ the Navier equations $\nabla\cdot\boldsymbol{\sigma}(\mathbf{u}) + \rho\,\omega^2\mathbf{u} + \mathbf{F}\delta = \mathbf{0}$ give, at order ω^2,

$$\nabla \cdot \boldsymbol{\sigma}(\mathbf{u}_1) = -\rho\, \mathbf{u}_0, \tag{22}$$

with no source: $\mathbf{u}_1$ and its traction are continuous at $z = z'$ and at $z = 0$. In each region the static field $\mathbf{u}_0$ is a polynomial times $\mathrm{e}^{\pm qz}$, the exponential being a double root of the static operator, so a particular solution of Eq. (22) is a polynomial of degree two higher times the same exponential. To it are added the static solutions $(a + bz)\mathrm{e}^{\pm qz}$ regular in the region. The whole-space problem is solved first; its ω^2 term, that of Kelvin's field, then forces the interface conditions that fix the reflected term $\mathbf{G}_1^R$ and the transmitted term $\mathbf{G}_1^T$. Every entry of $\mathbf{G}_1^R$ is $\mathrm{e}^{q(z+z')}$ times a polynomial in z and z', of degree at most two in each and three in all, with coefficients rational in q and in the Lamé constants and densities of the two media. The point propagator follows from $\mathbf{G}_1^R$ by the derivatives of Section 3, and its vertical moments over a cell are elementary.

Three checks confirm the result. When the media coincide, $\mathbf{G}_1^R = \mathbf{0}$ and $\mathbf{G}_1^T$ is the ω^2 term of Kelvin's field. Against the exact kernels, evaluated in 50-digit arithmetic, the residual $\mathbf{G} - \mathbf{G}_0 - \omega^2\mathbf{G}_1$ falls sixteenfold each time ω is halved, as an $O(\omega^4)$ remainder must. And the derivation done independently in two computer-algebra systems, one with symbolic and one with rational material constants, agrees to 5×10^{-51}.

In space the term needs more of the ladder. The divisions by q that turn a rotated entry into a lateral derivative (Section 3) take the lowest power of q in $\mathbf{G}_1^R$ to $q^{-5}\mathrm{e}^{-q\zeta}$, whose transform, like that of $q^{-4}\mathrm{e}^{-q\zeta}$, exists only as a finite part: the potentials Φ_3 and Φ_4 of Eq. (12). Since the finite part of a convergent integral is the integral, and the finite part is linear,

$$\mathbf{K}^R = \mathbf{K}_0^R + \omega^2\mathbf{K}_1^R + \frac{1}{4\pi^2}\,\mathrm{FP}\int_{\mathbb{R}^2}\big(\mathbf{Y} - \mathbf{Y}_0 - \omega^2\mathbf{Y}_1\big)\mathrm{d}^2k, \qquad \mathbf{K}_1^R = \frac{1}{4\pi^2}\,\mathrm{FP}\int_{\mathbb{R}^2}\mathbf{Y}_1\,\mathrm{d}^2k, \tag{23}$$

with $\mathbf{Y}$, $\mathbf{Y}_0$, $\mathbf{Y}_1$ the integrands of the block built from the dynamic kernel, $\mathbf{G}_0^R$ and $\mathbf{G}_1^R$, holds exactly, with no cutoff, when every finite part is taken by the same continuation in the power of

q.

The ω^2 block in closed form. Collected as in Section 3, the 9×9 ω^2 term of the reflected field is a sum of 744 terms $c\,z^A z'^B \partial^\alpha \Phi_j$ in 76 families, and that of the transmitted field of 582 terms in 61 families, with j up to 4 and the coefficients rational in the Lamé constants and densities of both media. In each, 81 terms in 16 families carry fewer lateral derivatives than their index j: Φ_1 with none, Φ_2 with at most one, Φ_3 with at most two and Φ_4 with at most three. For these the derivatives do not annihilate the solid harmonic of the residue, and the finite parts of Eq. (12) are needed as they stand. Their kernels are of degree $d = j - 1 - |\boldsymbol{\alpha}| \geq 0$, bounded at the singular point, but they are homogeneous only up to a logarithm. Under $\mathbf{x} \to \lambda\mathbf{x}$ the potential becomes $\Phi_j(\lambda\mathbf{x}) = \lambda^{j-1}\big[\Phi_j(\mathbf{x}) + \log\lambda\; g_0(\mathbf{x})\big]$, with $g_0 = -(-1)^n R^n P_n(\zeta/R)/(2\pi\, n!)$ minus the residue, so that every kernel $f = \partial^\alpha \Phi_j$ satisfies

$$f(\lambda\mathbf{x}) = \lambda^d\big[f(\mathbf{x}) + \log\lambda\; g(\mathbf{x})\big], \qquad \mathbf{x}\cdot\nabla f = d\,f + g, \qquad g = \partial^\alpha g_0, \tag{24}$$

g a homogeneous polynomial of degree d. It vanishes whenever the lateral derivatives annihilate the solid harmonic: for every term of the static field, and for 8 of the 16 finite-part families. Two of the steps of Section 4 change. Euler's identity on a box anchored at the singular point acquires a polynomial moment,

$$\big(3 + p + q + r + d\big)\int_{\text{box}} \rho_x^p \rho_y^q \zeta^r f \;=\; \sum_{\text{far faces}} |A_i| A_i^{n_i} \int_{\text{face}} \rho^{\mathbf{n}\setminus i} f \;-\; \int_{\text{box}} \rho_x^p \rho_y^q \zeta^r g, \tag{25}$$

and, since $d \geq 0$, its factor never vanishes. In a Duffy pyramid the kernel is $\rho^d[f + \log\rho\; g]$ on the far face, so the radial integrand gains a polynomial times $\rho^e \log\rho$, which is integrated exactly from the polynomial's coefficients in shifted Legendre polynomials and $\int_0^1 \rho^m \log\rho\,\mathrm{d}\rho = -1/(m+1)^2$. Finally, the scaling of a block with the cell size gains a term: in units of h a family contributes $h^{6+d+A+B}\big[\int f + \log h \int g\big]$. A finite part depends on the unit of length, here the kilometre, and the plane-wave remainder of Eq. (23), whose finite part at small q is taken in the same unit, depends on it in the opposite way, so that their sum does not. The line of images of Eq. (15), with its divergence at $t \to \infty$ dropped, is a different convention: it differs from Eq. (12) by polynomials, $-\gamma/2\pi$ and $\gamma\zeta/2\pi$ for $j = 1, 2$ and, for $j = 3, 4$, quadratics and cubics that are not even harmonic, so its closed forms are used only where the lateral derivatives annihilate the difference.

The closed form is checked against Eq. (23) itself. The radial integrand of $\mathrm{FP}\int \mathbf{Y}_1$, the azimuthal integral done, behaves as q^{-2} times an entire function at small q; on $[0, q_s]$ that function is sampled at Chebyshev nodes and each mode is integrated against q^{-2} exactly, the q^{-1} term giving $\log q_s$, and above q_s Gauss rules are used. For two reflected pairs whose images do not touch, over every field function and source monomial, the closed-form $\mathbf{K}_1^R$ and the finite part of its plane-wave integral agree to 5.1×10^{-14} and 6.0×10^{-14}, and for two transmitted pairs two half-widths apart across the interface the transmitted ω^2 block agrees to 5.4×10^{-14} and 3.6×10^{-14}; moving q_s from $h^{-1}/4$ to $h^{-1}/2$ changes the latter by about 10^{-13}.

The algebraic tail of touching cells. When both cells touch the interface the integrand of Eq. (23) decays only algebraically in q, and a plane-wave reference that truncates it is limited by its tail (Table 1). The tail is integrated analytically instead. With $c = qh$ the vertical moments of a cell are combinations of

$$E_k(c) = \int_{-1}^{1} v^k \mathrm{e}^{c(v-1)}\,\mathrm{d}v = P_k(1/c) + \mathrm{e}^{-2c} Q_k(1/c), \qquad P_k = \frac{1}{c} - \frac{k}{c}P_{k-1}, \quad Q_k = -\frac{(-1)^k}{c} - \frac{k}{c}Q_{k-1}, \tag{26}$$

with $P_0 = 1/c$, $Q_0 = -1/c$: polynomials in $1/c$, the first from the face of the cell nearest the interface and the second from the far face. A cell on the interface multiplies them by a prefactor of order one, any other cell by e^{-2qh} or less. The integrand is therefore

$$\mathbf{Y} = \mathbf{Y}_{\mathrm{alg}} + O(\mathrm{e}^{-2qh}), \qquad \mathbf{Y}_{\mathrm{alg}}(\mathbf{k}) = \sum_{p} \sum_{|n|\le 4} q^p\, \mathrm{e}^{\mathrm{i}n\phi}\, \mathbf{H}_{p,n}\, L_a(\mathbf{k})\, L_c(\mathbf{k})\, \mathrm{e}^{\mathrm{i}\mathbf{k}\cdot\boldsymbol{\Delta}}, \tag{27}$$

the product of the receiver's and the source's P parts with the frame kernel, rotated: finitely many powers (p from -8 to 1 for the blocks here), the nine harmonics of the rotation, and the exact coefficients $\mathbf{H}_{p,n}$. Each term is integrated over the whole plane in closed form. The transform of a power and a harmonic is the Weber–Schafheitlin integral,

$$\frac{1}{4\pi^2}\,\mathrm{FP}\int_{\mathbb{R}^2} q^p \mathrm{e}^{\mathrm{i}n\phi}\mathrm{e}^{\mathrm{i}\mathbf{k}\cdot\boldsymbol{\rho}}\,\mathrm{d}^2k = \frac{\mathrm{i}^n \mathrm{e}^{\mathrm{i}n\theta}}{2\pi}\,\frac{2^{p+1}\,\Gamma\!\left(\frac{|n|+p+2}{2}\right)}{\Gamma\!\left(\frac{|n|-p}{2}\right)}\,\rho^{-p-2} \tag{28}$$

(with a factor $(-1)^{|n|}$ for $n < 0$), continued in p: the convention of Eq. (12). The lateral moments and the phase turn it into an integral over the lateral separation $\mathbf{u} = \boldsymbol{\sigma} + \boldsymbol{\Delta}$ against the exact one-dimensional weights $w(\sigma) = \int P_E(v)(v-\sigma)^F \mathrm{d}v$, on the pieces $[\Delta-2, \Delta]$ and $[\Delta, \Delta+2]$ of each axis. On a piece with the origin at a corner, in polar coordinates about it, each monomial of the weights times the kernel is $\rho^d A(t)$, so that the radial integral is elementary and the angular one is a Gauss rule on a smooth integrand. Writing $p + \varepsilon$ for the power, the Gamma ratio, the radial integral $\int_0^R \rho^{d+1-\varepsilon}\mathrm{d}\rho$ and the scaling $h^{-\varepsilon}$ are each expanded in ε, and

$$\mathrm{FP} = \left[\varepsilon^0\right] \Gamma\text{-ratio}(\varepsilon) \times \int_{\mathrm{piece}} (\cdots)\,\rho^{-p-2-\varepsilon}(\varepsilon) \times h^{-\varepsilon}, \tag{29}$$

the constant term of the product. At most one factor has a pole: the Gamma ratio when the kernel is a polynomial in $\boldsymbol{\rho}$, and the radial integral when $d + 2 = 0$; the pole times the next order of the other factor gives the logarithms, and a zero of the Gamma ratio against a radial pole gives the local, delta-like terms. The remainder $\mathbf{Y} - \mathbf{Y}_{\mathrm{alg}}$ is integrated by Gauss rules and is below rounding beyond $qh = 24$.

The finite part at small q needs the same care. The part of $\mathbf{Y}_{\mathrm{alg}}$ below a split point q_s, which is subtracted from the whole-plane integral, cannot be sampled: it and the whole-plane integral are each about twenty times the block and cancel, and the powers down to q^{-7} amplify the rounding of any sampled representation, which left 3×10^{-10}. It is taken in closed form as well,

from the series of the Bessel function,

$$\mathrm{FP}\int_0^{q_s} q^{p+1} J_{|n|}(q\rho)\,\mathrm{d}q = \sum_{k\geq 0} \frac{(-1)^k(\rho/2)^{|n|+2k}}{k!\,(|n|+k)!} \begin{cases} \dfrac{q_s^{e+1}}{e+1}, & e \neq -1, \\ \log q_s, & e = -1, \end{cases} \qquad e = p+1+|n|+2k, \tag{30}$$

(the factor $(-1)^{|n|}$ for $n<0$ as in Eq. (28)), whose terms $\rho^{|n|+2k}\mathrm{e}^{\mathrm{i}n\theta}$ are polynomials in $\mathbf{u}$ and are integrated against the weights exactly. The full integrand below q_s, q^{-2} times an entire function, is integrated as before. The route shares no step with the closed form: no potentials Φ_j, no three-dimensional pieces. Table 2 compares the two for the static and the ω^2 blocks of touching cells. Each difference is matched by the change in the reference when q_s is halved, so that it measures the reference, not the closed form.

	static block		ω^2 block	
cells	closed vs. plane waves	q_s halved	closed vs. plane waves	q_s halved
cell and its own image	5.2×10^{-14}	3.8×10^{-14}	4.8×10^{-14}	1.1×10^{-13}
lateral neighbours, edge	7.5×10^{-14}	8.7×10^{-14}	1.3×10^{-13}	1.1×10^{-13}
lateral neighbours, corner	4.9×10^{-13}	4.3×10^{-13}	4.2×10^{-13}	2.1×10^{-13}
across the interface, face contact	3.4×10^{-14}	4.2×10^{-14}	3.9×10^{-14}	6.2×10^{-14}

Table 2. Blocks of cells touching the interface (4 field functions, 10 source monomials, 9×9): the closed forms (Sections 4 and 6) against the plane-wave integral with its algebraic tail in closed form, Eqs. (26)–(30), relative to the largest entry; and the change in the plane-wave integral when the split point q_s is moved from h^{-1} to $h^{-1}/2$.

The two corner routes, Duffy pyramids and Eq. (25), agree to 9×10^{-16} on the blocks of touching cells, and for identical media $\mathbf{K}_1^R$ vanishes. Until the closed form was available the term was used in the wavenumber domain, and that route is kept as the independent reference. With $\mathbf{Y}(\mathbf{k};\omega)$ the integrand of the reflected block, so that $\mathbf{K}^R = (4\pi^2)^{-1}\int \mathbf{Y}\,\mathrm{d}^2k$, and $\mathbf{Y}_0$, $\mathbf{Y}_1$ the integrands built in the same way from $\mathbf{G}_0^R$ and $\mathbf{G}_1^R$,

$$\mathbf{K}^R = \mathbf{K}_0^R + \omega^2\,\mathbf{K}_1^R(Q_0) + \frac{1}{4\pi^2}\int_{\mathbb{R}^2}\left[\mathbf{Y}-\mathbf{Y}_0-\omega^2\,\mathbf{Y}_1\,\theta(q-Q_0)\right]\mathrm{d}^2k, \qquad \mathbf{K}_1^R(Q_0) = \frac{1}{4\pi^2}\int_{q>Q_0}\mathbf{Y}_1\,\mathrm{d}^2k. \tag{31}$$

Neither $\mathbf{K}_0^R$ nor $\mathbf{K}_1^R(Q_0)$ depends on the frequency: both are computed once for each arrangement of cells, the first in closed form (Section 4) and the second as a plane-wave integral of a closed-form integrand, which needs no reflection coefficients. Only the last integral is evaluated at each frequency, and beyond Q_0 its integrand falls by two more powers of q than with the static image alone.

Table 4 shows the effect for the block of Table 1, at $\omega h/\beta = 1/6$ and at $\omega h/\beta = 1$, the frequency at which a shear wavelength spans about three cells. The remainder now converges as Q^{-6}. Truncated at $Qh = 10$ it leaves 3.6×10^{-10} of the block at the lower frequency, where the static subtraction alone leaves 1.3×10^{-6} and needs $Qh \approx 160$ to do as well; at $\omega h/\beta = 1$ it leaves 3×10^{-7} at $Qh = 10$ and 6×10^{-11} at $Qh = 40$, where the static subtraction alone leaves 3×10^{-5} and 1×10^{-7}. The transmitted coupling behaves in the same way. For a source cell touching the interface and the field cell its mirror image in B, so that the two touch across the interface, the

shells of $\mathbf{Y} - \mathbf{Y}_0$ fall with local orders 3.90 to 4.17 and those of $\mathbf{Y} - \mathbf{Y}_0 - \omega^2\mathbf{Y}_1$ (with $\mathbf{Y}_1$ built from $\mathbf{G}_1^T$) with orders 5.86 to 6.24; truncated at $Qh = 10$ the remainder leaves 2.2×10^{-10} of the block at $\omega h/\beta = 1/6$, and truncated at $Qh = 40$ it leaves 4.3×10^{-11} at $\omega h/\beta = 1$.

The lower cutoff also settles the precision of the mode sums. Below $Qh = 10$ the cancellation between the P and SV terms (Section 2) is mild: evaluated in double precision, the integral over $qh < 10$ differs from its 40-digit value by 7×10^{-12} of the block at $\omega h/\beta = 1/6$ and by 5×10^{-13} at $\omega h/\beta = 1$, and the shell $10 < qh < 20$ by 6×10^{-11} and 2×10^{-14}. With the ω^2 term subtracted, a double-precision plane-wave integral therefore gives the reflected block of a touching cell to about 4×10^{-10} with the cutoff at $Qh = 10$ when $\omega h/\beta = 1/6$, and to about 10^{-10} with the cutoff at $Qh = 40$ when $\omega h/\beta = 1$; with the static image alone the same accuracy needs cutoffs at which double precision fails. The frequency-independent integral $\mathbf{K}_1^R(Q_0)$ is the slowly convergent one, with an integrand that falls as $\mathbf{Y} - \mathbf{Y}_0$ does. Its integrand needs no mode sum: written in the frame of $\mathbf{k}$ it is $\mathrm{e}^{q(z+z')}$ times a polynomial in z and z' whose coefficients are Laurent polynomials in q, and in double precision it agrees with the 40-digit kernel to 10^{-12}, an error that enters multiplied by ω^2. It is not integrated to infinity: by Eq. (23), $\mathbf{K}_1^R(Q_0) = \mathbf{K}_1^R - (4\pi^2)^{-1}\mathrm{FP}\int_{q<Q_0} \mathbf{Y}_1 \, \mathrm{d}^2k$, the closed-form block less the finite part below Q_0, so that no tail is left. (Integrated by octaves to $Q_1 h = 160$ with the tail taken from the q^{-4} law of the last octave, as before the closed form existed, it carried 5×10^{-7} of itself; by octaves to $Q_1 h = 640$, 8×10^{-10}.) The angular dependence costs little: the rotation of the frame kernel to the direction of $\mathbf{k}$ contains only the harmonics $\mathrm{e}^{in\phi}$ with $|n| \le 4$, so at each q the azimuthal rule only sums the cells' lateral moments against nine harmonics.

Assembled in this way, with the plane-wave integral below Q_0 in double precision, the split reproduces the reference values to the size of the remainder it drops. For identical media the transmitted block must be the touching block of the homogeneous medium, which Nestor [1] obtains in closed form with no plane waves; over every field function and source monomial ($4 \times 10 \times 9 \times 9$ entries), for cells touching across the interface face to face, along an edge and at a corner, and with the source above or below, the two agree to 0.8 to 3.2×10^{-10} at $\omega h/\beta = 1/6$ with $Q_0 h = 10$ and to 1.7×10^{-12} to 5.4×10^{-11} at $\omega h/\beta = 1$ with $Q_0 h = 40$. For different media the reflected block of a cell with its own image and the transmitted block of two cells touching across the interface agree with the 40-digit integrals of Tables 1 and 4 to 3.6×10^{-10} and 2.2×10^{-10} at the lower frequency and to 5.9×10^{-11} and 3.9×10^{-11} at the higher, the remainders of Table 4 and of the transmitted sweep.

The integral below Q_0, the only part evaluated at each frequency, is cheapest in the same frame. Its integrand is the dynamic frame kernel, rotated, times the cells' lateral moments and phase; the rotation holds nine harmonics, so the integral is a sum over radial nodes of the harmonics of the frame kernel, one 6×6 interface solve per node with the cells' vertical moments in closed form, against the azimuthal integrals $S_n(q)$ of the lateral moments and phase. These depend on neither the frequency, the media nor the depths of the cells: they are tabulated once for each lateral offset, as piecewise Chebyshev series, and interpolated to the radial rule of each frequency, whose panel ends follow the branch points $q = \omega/c$. The result agrees with the mode sum evaluated at every point of the polar grid to 3×10^{-15} to 2×10^{-11} (the latter the rounding of the reflected mode sums, which both routes share) and costs a twentieth to a fiftieth of its time.

The next order. The perturbation of the static problem continues: at order ω^4, $\nabla\cdot\boldsymbol{\sigma}(\mathbf{u}_2) = -\rho\,\mathbf{u}_1$ with no source, and $\mathbf{u}_2$ is a polynomial two degrees higher than $\mathbf{u}_1$ times the same exponential, plus the regular homogeneous solutions; Kelvin's term and the reflected and transmitted terms $\mathbf{G}_2^R$, $\mathbf{G}_2^T$ follow from the conditions at z' and at the interface as before. Against the exact kernels at 50 digits the residual $\mathbf{G}-\mathbf{G}_0-\omega^2\mathbf{G}_1-\omega^4\mathbf{G}_2$ falls 64-fold each time ω is halved, and for identical media $\mathbf{G}_2^R=\mathbf{0}$ and $\mathbf{G}_2^T$ is Kelvin's ω^4 term. Derived with the Lamé constants and densities as symbols, each entry is the exponential times a polynomial in q, z, z' and the media over q^5 and a product of μ_A, μ_B, $\lambda_A+2\mu_A$, $\lambda_B+2\mu_B$, $\mu_A+\mu_B$ and the two factors coupling the media, $\lambda_A(\mu_A+\mu_B)+\mu_A^2+3\mu_A\mu_B$ and $\lambda_B(\mu_A+\mu_B)+3\mu_A\mu_B+\mu_B^2$; at rational media it agrees identically with an independent derivation in a second computer-algebra system. Its lowest power of q in the frame is q^{-5}, and collected as in Section 3 the 9×9 term is a sum of 1245 (reflected) and 1002 (transmitted) canonical terms with potentials up to Φ_6, about 550 of them finite parts, each coefficient a polynomial in z, z' with coefficients rational in the media. Over the common denominator the collection is polynomial arithmetic on the numerators alone, and the factors of the denominator are cancelled by exact division at the end. Their closed-form block $\mathbf{K}_2$ is formed as $\mathbf{K}_1$ is, and the split becomes

$$\mathbf{K}=\mathbf{D}(\omega;Q_0)+\mathbf{A}_0+\omega^2\mathbf{A}_1+\omega^4\mathbf{A}_2+\mathbf{E},\qquad \mathbf{A}_n=\mathbf{K}_n-\frac{1}{4\pi^2}\,\mathrm{FP}\int_{q<Q_0}\mathbf{Y}_n\,\mathrm{d}^2k, \tag{32}$$

$\mathbf{D}$ the plane-wave integral of $\mathbf{Y}$ below Q_0 at each frequency and $\mathbf{E}=(4\pi^2)^{-1}\int_{q>Q_0}(\mathbf{Y}-\mathbf{Y}_0-\omega^2\mathbf{Y}_1-\omega^4\mathbf{Y}_2)\mathrm{d}^2k$ the dropped remainder, which now falls as Q_0^{-8}. Each $\mathbf{A}_n$ is the part of the closed-form block beyond Q_0, obtained without integrating any tail. For $\mathbf{A}_2$ the subtraction cancels about seven digits, since the part of $\mathbf{K}_2$ beyond Q_0 is about 10^{-7} of it: the finite part below Q_0 must be as accurate as the closed form.

The finite part at small q in closed form. Near $q=0$ the integrands $\mathbf{Y}_n$ behave as q^{-2} $(n=0,1)$ and q^{-4} $(n=2)$ times entire functions. A finite part taken from samples of the entire function, at Chebyshev nodes with each mode integrated exactly, is limited by the rounding of the samples, which the factor q_s^{1-N} amplifies: for q^{-4} it has a narrow window, 6×10^{-9} of $\mathbf{K}_2$ with 24 nodes on $[0,3/h]$, 4×10^{-11} with 40 and 5×10^{-10} with 48. It is taken instead in closed form, with every factor of the integrand an exact power series in q: the frame kernel's Laurent polynomial; the vertical moments,

$$\int_{\mathrm{cell}} f(v)\,z^a\mathrm{e}^{sqz}\,\mathrm{d}z=\sum_{n\ge0}\frac{(sq)^n}{n!}\int_{\mathrm{cell}} f(v)\,z^{a+n}\,\mathrm{d}z, \tag{33}$$

exact polynomial moments; and the lateral moments with the phase, for each harmonic, by the series of the Bessel function against the exact lateral weights, Eq. (30). Their product is a Laurent series $\sum_p q^p\,\mathbf{R}_{p,n}$ for each harmonic, and each term's finite part on $[0,q_s]$ is a power of q_s, or $\log q_s$ at the power -1. The series converge for every q, but lose digits as q_s times the largest distance R within the pair grows, so $q_s=\min(1/h,4/R)$; above q_s Gauss rules are used. At order zero, whose integrand is integrable at $q=0$, the closed form agrees with plain integration to 1.6×10^{-15} and 1.4×10^{-14}; and $\mathbf{K}_2$ less the closed-form finite part below Q_0 agrees with the

integral of $\mathbf{Y}_2$ beyond Q_0 by octaves (whose shells fall as q^{-6}) to 1.9×10^{-15} (reflected) and 1.2×10^{-15} (transmitted) of $\mathbf{K}_2$. With it, Eq. (32) reproduces the 40-digit references to the size of $\mathbf{E}$ (Table 3): a factor of about 2^8 per doubling of Q_0, and below 10^{-12} at $Q_0 h = 40$ when $\omega h/\beta = 1$.

	$\omega h/\beta = 1/6$		$\omega h/\beta = 1$		
	$Q_0 h = 10$	20	$Q_0 h = 10$	20	40
reflected, to ω^2	3.6×10^{-10}	2.9×10^{-11}	3.1×10^{-7}	4.8×10^{-9}	6.3×10^{-11}
reflected, to ω^4	1.1×10^{-11}	2.6×10^{-11}	3.1×10^{-9}	1.2×10^{-11}	8.8×10^{-13}
transmitted, to ω^2	2.2×10^{-10}	3.5×10^{-12}	2.0×10^{-7}	3.2×10^{-9}	4.2×10^{-11}
transmitted, to ω^4	6.3×10^{-14}	1.2×10^{-14}	1.6×10^{-9}	6.1×10^{-12}	8.1×10^{-14}

Table 3. The block of a cell with its own image (reflected) and of two cells touching across the interface (transmitted) by the split Eq. (32), against the 40-digit plane-wave references, relative to the block: with the asymptotic terms to ω^2 and to ω^4, the finite parts below Q_0 in closed form. The reflected entries at $\omega h/\beta = 1/6$ with ω^4 are the rounding of the double-precision mode sums below Q_0, whose P and SV terms cancel at that frequency.

	$\omega h/\beta = 1/6$				$\omega h/\beta = 1$			
	less $\mathbf{Y}_0$		less $\mathbf{Y}_0 + \omega^2 \mathbf{Y}_1$		less $\mathbf{Y}_0$		less $\mathbf{Y}_0 + \omega^2 \mathbf{Y}_1$	
shell qh	contr.	order	contr.	order	contr.	order	contr.	order
10–20	1.2×10^{-6}		3.5×10^{-10}		3.0×10^{-5}		3.1×10^{-7}	
20–40	7.7×10^{-8}	4.00	5.5×10^{-12}	6.00	1.8×10^{-6}	4.01	4.7×10^{-9}	6.01
40–80	4.3×10^{-9}	4.17	7.2×10^{-14}	6.25	1.0×10^{-7}	4.18	6.2×10^{-11}	6.25
80–160	2.8×10^{-10}	3.90	1.2×10^{-15}	5.87	6.8×10^{-9}	3.90	1.1×10^{-12}	5.87
160–320	1.8×10^{-11}	4.01	1.9×10^{-17}	6.03	4.2×10^{-10}	4.01	1.6×10^{-14}	6.02

Table 4. The block of Table 1 with the static image subtracted, and with the static image and the ω^2 term subtracted, at two frequencies: the contribution of each shell of the lateral wavenumber, relative to the block, and the local order. All kernels in 40-digit arithmetic; halving the radial panels and doubling the azimuthal rule changes the shell 160–320 of the second remainder by 8×10^{-20} and 5×10^{-20} of the block.

7 Verification

Each step was checked against a route that shares no step with it; Table 5 collects the results. The media are A with $\alpha = 5.0$, $\beta = 3.0\,\mathrm{km\,s^{-1}}$, $\rho = 2.5\,\mathrm{g\,cm^{-3}}$ and B with $\alpha = 6.5$, $\beta = 3.6\,\mathrm{km\,s^{-1}}$, $\rho = 2.8\,\mathrm{g\,cm^{-3}}$, and $h = 0.05\,\mathrm{km}$. Errors are relative to the largest entry of the quantity compared, and the corner moments are measured on the scale 2^{p+q+r} of each monomial.

The checks are of four kinds. The spectral solutions were derived independently in two computer-algebra systems, and their spatial forms were compared with direct two-dimensional Fourier integration of the spectral kernels. The corner moments were compared with three routes: Euler's reduction with Gauss rules on the far faces, tanh-sinh quadrature of the singular volume integral itself, and an evaluation at thirty digits that integrates one coordinate exactly and then the other two with Duffy coordinates at the corner [10]. We note that adaptive three-dimensional

cubature applied directly with the singular point at a corner, or over the semi-infinite column, can report convergence while in error by as much as 5×10^{-7}, which is why the references integrate one coordinate exactly. The blocks of cells away from the singular point were compared with a tensor Gauss rule over both cells of the canonical kernel, which shares neither the separation form nor the exact weights. Two limits close the chain: for identical media the transmitted blocks must be the touching blocks of the homogeneous medium [1], computed by an unrelated method, and the image must vanish; and the touching block of a cell with its own image must be the limit of the plane-wave integral (6) of the static spectrum as the cutoff grows.

quantity	compared with	agreement
Plane-wave representation		
integrand of Eq. (2)	residue form of the plane-to-plane kernel	3×10^{-15}
blocks (3), planes not touching	closed-form blocks [1]	2–6×10^{-13}
Static image		
reflected spectral tensor	second computer-algebra derivation	5×10^{-16}
dynamic reflected integrand of a touching block, $qh = 20$–1280	static integrand	$\approx 2(\omega/\beta q)^2$
spatial form	Fourier integral of the spectrum	7×10^{-13}–1.3×10^{-12}
canonical terms (14)	direct differentiation of the spatial kernel	3×10^{-15}
Corner moments		
Euler's reduction, 7 cases, up to 4 derivatives	tanh-sinh quadrature of the volume integral	6×10^{-16}–4.8×10^{-14}
closed forms, all 53 352 moments, monomials to degree $(6, 6, 8)$	Euler's reduction with Gauss rules on the faces	1.2×10^{-15}
13 moments, all potentials, 3 degenerate columns	thirty-digit evaluation	5×10^{-23}–1.6×10^{-16}
B_1, B_2, Eq. (20)	forty-digit angular quadrature	10^{-40}
Finite-part potentials		
degree derivative (11), $n \le 5$	numerical derivative in ν	2×10^{-41}
Φ_1–Φ_4, Eq. (12)	finite part of the integral (10) itself, 40 digits	3×10^{-40}
Φ_1–Φ_6: ladder, harmonicity	symbolic	exact
Reflected blocks		
cells away from the singular point, 4 pairs, degree-one and degree-two fields	tensor Gauss rule over both cells	3×10^{-15}–8.5×10^{-15}
touching blocks, closed-form corner moments, 3 pairs	Gauss rules on the far faces	1.8–5.3×10^{-15}
cell with its own image	static plane-wave integral, $Qh = 320$ to $20\,480$	$7.9 \times 10^{-6} \to 2.3 \times 10^{-8}$
Transmitted field and blocks		

continued

Table 5, continued

quantity	compared with	agreement
transmitted spectral tensor	second computer-algebra derivation	4.2×10^{-16}
spatial form	Fourier integral of the spectrum	3.9×10^{-16}
canonical terms, different media	Fourier integral of the spectrum	6.6–7.8×10^{-13}
identical media: face, edge, corner contact; mirrored; degree two	homogeneous-medium blocks [1]	1.9–7.6×10^{-15}
different media, cells not touching	tensor Gauss rule over both cells	1.7–5.0×10^{-13}
Order ω^2 and the split (31)		
$\mathbf{G}_1^R$, $\mathbf{G}_1^T$	exact kernels at 50 digits, ω halved	residual falls 16-fold
$\mathbf{G}_1^R$, $\mathbf{G}_1^T$	second computer-algebra derivation	5×10^{-51}
frame integrands $\mathbf{Y}_0$, $\mathbf{Y}_1$ in double precision, $qh = 0.5$–160	40-digit kernels	2×10^{-14}, 1.2×10^{-12}
identical media, touching across the interface, 3 contacts, mirrored	homogeneous-medium blocks [1]	1.7×10^{-12}–3.2×10^{-10}
different media, reflected and transmitted touching blocks	40-digit plane-wave integrals	3.9×10^{-11}–3.6×10^{-10}
ω^2 terms (14), 744 and 582, reflected and transmitted	transform of each term against the rotated, differentiated spectrum, 30 digits	exact
finite-part kernels: Eq. (24), 16 families	direct evaluation, 30 digits	exact
corner boxes, Eq. (25) and Duffy pyramids, 26 moments, 9 families, Φ_0–Φ_4, $d = -3$ to 1, with and without g	25-digit Duffy quadrature, tanh-sinh in ρ	8.3×10^{-16}, 4.3×10^{-16}
ω^2 blocks, two reflected pairs not touching their images, two transmitted pairs	finite part of the plane-wave integral of $\mathbf{Y}_1$	3.6–6.0×10^{-14}
$\mathbf{G}_2^R$, $\mathbf{G}_2^T$	exact kernels at 50 digits, ω halved	residual falls 64-fold
$\mathbf{G}_2^R$, $\mathbf{G}_2^T$ with symbolic media	second computer-algebra derivation at rational media	identical
ω^4 terms, 1245 and 1002, symbolic media	transform of each term against the rotated, differentiated spectrum, 30 digits	exact
ω^4 blocks, two pairs of media, two reflected pairs not touching their images, two transmitted pairs	finite part of the plane-wave integral of $\mathbf{Y}_2$: closed form below q_s, Gauss panels above	4.6×10^{-16}–2.9×10^{-15}
Galerkin solve, 8 cells straddling the interface, identical media	the homogeneous solve	3.1×10^{-12}

continued

Table 5, continued

quantity	compared with	agreement
Galerkin solve, different media, scattered field at 4 receiver cells, vanishing contrast	Born field, every coupling by the plane-wave integral of the interface coefficients	4.9–5.6×10^{-11}
lateral radius $16h$ of the corrections, C^∞ window, row of 28 cells	full windowed set-up	8.3×10^{-11} (solution)
GMRES with lateral FFT products, $2 \times 5 \times 5$ cells, 10% contrast	dense solve from the same blocks	4.6×10^{-12}
depth radius $4h$ of the corrections, six layers of two cells	full windowed set-up	1.8×10^{-10} (blocks), 2.7×10^{-11} (solution)
$\mathbf{D}$ of touching cells at 8–$64h$ by the frame route	polar sum over the modes	$\leq 7 \times 10^{-12}$ of the touching block
$\mathbf{D}_{\text{high}}$ on the wedge by the lateral symmetries	the whole periodic grid	2.4×10^{-16}
$\mathbf{D}_{\text{low}} + \mathbf{D}_{\text{high}}$, $2 \times 4 \times 4$ cells, every class	unsplit $\mathbf{D}$	1.3×10^{-10}, 2.9×10^{-9} of the touching block
mode factors of $\mathbf{D}$'s integrand, reflected and transmitted, on the ramp	the integrand point by point	rounding of the modes' cancellation
whole-space march, planes ≥ 3 apart (planes 0 to 2 apart stored), $\omega h/\beta_A = 1/6,\ 1/2$	Paper 2's blocks	3.4×10^{-10}, 1.6×10^{-9}
coupling product without stored kernels (lattice nodes by FFT, tuned low band), 50 and 96 cells; $8 \times 3 \times 3$, the march acting	stored-kernel product	1.7×10^{-11}; 1.5×10^{-11}
two-way march preconditioner through the recursions, $8 \times 2 \times 2$ cells, $\omega h/\beta_A = 1/6,\ 1/2$	the same on the stored kernels of every pair of planes	1.5×10^{-12}, 1.9×10^{-11}
Algorithm 5.1, Dirichlet-to-Neumann form, one plane, contrast varying	the plane's system	2.8×10^{-14} (inverse)
its window operator over all planes, 15 cells in 4 planes	true system on the padded grid, assembled from the stored kernels	5.3×10^{-16}
FGMRES with the march, Algorithm 5.1 averaged, DtN ($w = 0$, all), 15 cells, contrast varying	dense solve	1.6, 9.7, 8.3, 7.8×10^{-12}
split to ω^2 and to ω^4 at $Q_0 h = 3$–10, self-image, across and lateral pairs, $\omega h/\beta = 1/6,\ 1/2$	split to ω^4 at $Q_0 h = 20$	at $Q_0 h = 10$: 3.5×10^{-10}, 2.5×10^{-8} (ω^2); 4.8×10^{-11}, 6.2×10^{-11} (ω^4)
$\mathbf{K}_2$ less the finite part below Q_0 in closed form	octaves of $\mathbf{Y}_2$ beyond Q_0	1.2–1.9×10^{-15} of $\mathbf{K}_2$
finite part at small q in closed form, order zero	plain integration (integrable)	1.4×10^{-14}

continued

Table 5, continued

quantity	compared with	agreement
static and ω^2 blocks of touching cells: own image, edge, corner, across the interface	plane-wave integral, algebraic tail in closed form (Weber–Schafheitlin, 2-D lateral finite parts)	3.4×10^{-14}–4.9×10^{-13}
split with a smooth window, Eq. (34), rows of 5 cells, reflected and transmitted	split with the sharp cutoff, pair by pair	3.9×10^{-14}–1.9×10^{-11}
azimuthal integrals S_n by Eq. (35), offsets to $40h$	direct azimuthal rule	4×10^{-16}–3×10^{-13}

Table 5. Independent checks. The last entry of each group of blocks is limited by the accuracy of the comparison route: the Gauss rule over both cells at order 10 to about 10^{-13}, and the static plane-wave integral by its algebraic convergence; those of the split are the remainder it drops, at $Q_0h = 10$ ($\omega h/\beta = 1/6$) and $Q_0h = 40$ ($\omega h/\beta = 1$).

The accuracy of the touching blocks themselves is measured against a reference that removes the rounding of the double-precision routes: the power-basis route carried out in 80-bit extended arithmetic (unit round-off 1.1×10^{-19}), with the corner moments in closed form at 40 digits, Gauss rules refined to extended precision, and the kernels and exact weights rounded to extended precision only. Its own rounding, bounded by the unit round-off times the sum of the magnitudes of its terms, is below 6×10^{-17} of the blocks. Against it (Table 7) the well-conditioned evaluation is at round-off for both field degrees, 3×10^{-16} to 1.2×10^{-15}, and the power basis is the same at degree one but loses up to a digit at degree two. The fully independent check of these blocks, the plane-wave integral, is limited to 2.3×10^{-8} by its own convergence. With identical media the image vanishes identically, to the last bit, on both sides of the interface.

The plane-wave route, which shares no step with the closed forms, has to reach every pair of cells near the interface, not only the touching ones. Two devices make it do so; both belong to the reference, not to the method. First, the cutoff $\theta(q-Q_0)$ of Eq. (31) makes the frequency-independent corrections $\mathbf{K}_0^R - (4\pi^2)^{-1}\int_{q<Q_0}\mathbf{Y}_0$ and $\mathbf{K}_1^R(Q_0)$ ring in space, so that they decay with the cells' lateral offset d only as about $d^{-3/2}$: for a row of cells touching the interface at $\omega h/\beta = 1/6$ and $Q_0h = 10$ the first is still 1.4×10^{-5} of the touching block at $d = 32h$. A smooth window $\chi(q)$, zero below $Q_0/4$, one above Q_0, and $\mathrm{e}^{-1/t}/(\mathrm{e}^{-1/t}+\mathrm{e}^{-1/(1-t)})$ on the ramp between (t the position on the ramp), moves the ramp integrals from the plane-wave part to the corrections,

$$\mathbf{K}^R = \frac{1}{4\pi^2}\int_{q<Q_0}\big[\mathbf{Y}-\chi(\mathbf{Y}_0+\omega^2\mathbf{Y}_1)\big]\mathrm{d}^2k+\Big(\mathbf{K}_0^R-\frac{1}{4\pi^2}\int(1-\chi)\,\mathbf{Y}_0\,\mathrm{d}^2k\Big)+\frac{\omega^2}{4\pi^2}\int\chi\,\mathbf{Y}_1\,\mathrm{d}^2k+\mathbf{E}, \quad (34)$$

with the same remainder $\mathbf{E}$ beyond Q_0; the corrections then decay faster than any power of d, to 1.2×10^{-6}, 5.9×10^{-9}, 9.1×10^{-11} and 4.7×10^{-12} of the touching block at $d = 8h$, $16h$, $24h$ and $32h$, reflected and transmitted alike (Table 6). The window costs nothing: the corrections are the sharp ones plus integrals of $\chi\mathbf{Y}_0$ and $\chi\mathbf{Y}_1$ over the ramp, below Q_0, which the plane-wave part gives back, and the windowed and sharp blocks agree to the rounding of the plane-wave part. The wider the ramp, the faster the decay: a ramp from $Q_0/2$ gives 3.2×10^{-11} at $32h$ instead of 4.7×10^{-12}, and a polynomial step, smooth only to finite order, decays algebraically and stalls

near 10^{-10}. The ω^2 correction decays alike and is two to three orders smaller at this frequency: 2.7×10^{-11} of the touching block at $16h$. So the corrections are local: beyond $12h$, $16h$ and $24h$ (six, eight and twelve cells) they fall below 10^{-6}, 10^{-8} and 10^{-10} of the touching block, and the block is the windowed plane-wave integral below Q_0 alone. Within that radius they depend only on the two depths and the lateral offset, so a row or grid of cells along the interface needs them once per pair of depths and offset, and the costly ω^2 part only within its own, smaller radius.

window d/h	sharp $\mathbf{A}_0$	C^∞ from $Q_0/2$ $\mathbf{A}_0$	C^∞ from $Q_0/4$ $\mathbf{A}_0$	C^∞ from $Q_0/4$ $\omega^2\mathbf{A}_1$	polynomial from $Q_0/2$ $\mathbf{A}_0$
4	2.7×10^{-4}	9.8×10^{-5}	1.2×10^{-4}	2.7×10^{-7}	7.9×10^{-5}
8	1.0×10^{-4}	1.7×10^{-6}	1.2×10^{-6}	5.1×10^{-9}	5.4×10^{-7}
16	3.2×10^{-5}	1.5×10^{-8}	5.9×10^{-9}	2.7×10^{-11}	9.8×10^{-9}
24	2.4×10^{-5}	5.5×10^{-10}	9.1×10^{-11}	4.4×10^{-13}	8.8×10^{-10}
32	1.4×10^{-5}	3.2×10^{-11}	4.7×10^{-12}	3.8×10^{-14}	1.4×10^{-10}

Table 6. Locality of the frequency-independent corrections: a row of cells touching the interface at lateral offset d from a touching cell, full $4 \times 10 \times 9 \times 9$ blocks, $Q_0 h = 10$, $\omega h/\beta = 1/6$, the largest entry relative to the largest of the touching static block. Reflected; transmitted within a factor 1.3. With the sharp cutoff the decay is about $d^{-3/2}$; with the C^∞ window, faster than any power; with the polynomial step $I_t(4, 4)$, algebraic.

Second, the azimuthal integrals of the cells' lateral moments $L_a L_c$ against the phase $\mathrm{e}^{\mathrm{i}\mathbf{k}\cdot\boldsymbol{\Delta}}$ of a lateral offset $\boldsymbol{\Delta}$ and the nine harmonics of the rotation need no rule that grows with $q|\boldsymbol{\Delta}|$: by the Jacobi–Anger expansion of the phase,

$$S_n(q; \boldsymbol{\Delta}) = \frac{1}{2\pi} \sum_p \mathrm{i}^p J_p(q|\boldsymbol{\Delta}|)\, \mathrm{e}^{-\mathrm{i}p\phi_\Delta}\, c_{-n-p}(q), \tag{35}$$

where $c_m(q)$ are the azimuthal Fourier coefficients of $L_a L_c$. These are band-limited by the cell, not by the offset ($|m| \le 13$, 51 and 148 above 10^{-12} at $qh = 1$, 10 and 40), so they are computed once for every offset.

cells	field	power basis	well-conditioned
reflected, cell with its own image	degree one	9.2×10^{-16}	8.2×10^{-16}
reflected, lateral neighbour	degree one	6.8×10^{-16}	7.3×10^{-16}
reflected, diagonal neighbour	degree one	6.6×10^{-16}	4.0×10^{-16}
transmitted, face contact	degree one	1.2×10^{-15}	1.2×10^{-15}
transmitted, corner contact	degree one	3.4×10^{-16}	4.0×10^{-16}
reflected, cell with its own image	degree two	9.6×10^{-15}	9.2×10^{-16}
reflected, diagonal neighbour	degree two	1.7×10^{-15}	6.5×10^{-16}
transmitted, face contact	degree two	9.6×10^{-15}	4.1×10^{-16}

Table 7. The touching blocks against the extended-precision reference: for each kernel family, the largest error over the field and source functions relative to the largest entry, and the largest over the families. Degree one: 4 field functions and 10 source monomials; degree two: 10 and 35.

The blocks in a solve. The blocks enter the Galerkin system of the homogeneous medium unchanged in form, $(\mathbf{M} - \mathbf{K}\mathbf{E})\boldsymbol{\psi} = \langle \mathbf{Q}, \mathbf{u}_{\mathrm{ref}} \rangle$ [1, 5], with the coupling of the two half-spaces: for

cells on one side the whole-space block of that side's medium plus the reflected block, for cells on opposite sides the transmitted block alone. Each cell's contrast is taken against its own side's medium, and the right-hand side is the field of the two half-spaces for an incident plane wave, the incident and reflected waves above the interface and the transmitted ones below it. Eight cells straddle the interface, four on each side, each with a contrast of 10% of its medium's λ, μ and ρ; $h = 50\,\mathrm{m}$, $\omega h/\beta_A = 1/6$, and a P wave with lateral slowness $(0.3, 0.1)/\alpha_A$. With identical media the solve reproduces the homogeneous one, assembled from the whole-space blocks alone with the incident wave as right-hand side, to 3.1×10^{-12}. With different media, four receiver cells without contrast, two above the interface and two below, in planes clear of the scatterer's, carry the scattered field; solved for directly, $(\mathbf{M} - \mathbf{KE})\mathbf{s} = \mathbf{KE}\boldsymbol{\psi}_0$, and taken to vanishing contrast, it agrees to 4.9–5.6×10^{-11} with the Born field whose every coupling, direct, reflected or transmitted, is the plane-wave integral of the mode reflection and transmission coefficients of the interface, sharing no step with the closed forms or the split. The scattered field at the receivers changes by 8.1×10^{-3} from cells of degree zero to degree two and by 2.2×10^{-4} from degree one to two. The frequency-independent parts are computed once for each class of pairs under the lateral symmetries of the interface (78 classes, about 800 s on four processes); each further frequency then costs 3.5 s for all the blocks and a fraction of a second for the solve.

Larger scatterers. Every block depends only on the two depth layers and the lateral offset, so for each pair of layers the coupling of a grid of cells is a lateral convolution. Its blocks over the offsets of the grid's lateral box are transformed once per frequency, with zero padding to $2n - 1$ for n cells; a product then costs one two-dimensional FFT per source layer, one inverse per receiver layer and a contraction per pair of layers, and no block of a single pair is formed. GMRES solves the system with these products, preconditioned block by block with $(\mathbf{M} - \mathbf{K}_{mm}\mathbf{E}_m)^{-1}$, $\mathbf{K}_{mm}$ the self block with its own image. The set-up keeps the corrections only within the radius $16h$ of Table 6, with the C^∞ window from $Q_0/4$, which moves the solution by 8.3×10^{-11} on a row of 28 cells. On a box of $2 \times 5 \times 5$ cells straddling the interface, each with a contrast of 10% of its medium, in the media, frequency and incident wave above, at degree one, GMRES with relative tolerance 10^{-11} takes 8 products and agrees with the dense solve from the same blocks to 4.6×10^{-12}. On $2 \times 12 \times 12$ cells (288) it takes 9 products and 0.7 s, so the iteration count barely grows with the grid at this contrast. The cost is elsewhere: the set-up takes 1477 s on four processes (312 classes, 128 of them within the radius; 364 s for the 60 classes of the smaller box), and the transformed kernels of one frequency take 183 s and 110 MB (92 s and 17 MB). Little of the set-up is the closed forms: the tables S_n of the lateral harmonics take 43% of it, and the cost of a table grows as the square of its lateral reach. A table depends on the offset alone, so one per offset suffices, not one per pair of layers.

Locality in depth. The corrections are local in depth as well. For two cells at lateral offset zero with a total gap $g = d_r + d_s$ between them and the interface, $|\mathbf{A}_0 + \omega^2\mathbf{A}_1|$ falls to 2.4×10^{-2}, 2.6×10^{-6}, 3.7×10^{-9} and 3.7×10^{-12} of the touching static block at $g = 0$, $2h$, $4h$ and $6h$. This is about 10^{-4} per $2h$, alike for reflection on either side and for transmission: the C^∞ step is negligible early on its ramp, so the window effectively opens near $Q_0/2$. Beyond $g = 4h$ the corrections fall below 10^{-8}, and only the three layers nearest the interface on each side carry

them, whatever the number of layers. On six layers of two cells, dropping them from $g = 4h$ keeps the corrections of 24 of the 72 classes and moves the blocks by 1.8×10^{-10} of the largest and the solution by 2.7×10^{-11}; the set-up then takes 175 s instead of 469 s.

The plane-wave part on the FFT grid. With the corrections bounded in both directions, the plane-wave part $\mathbf{D}$ is the cost of each frequency, and it is not local. For two touching cells it is still 5.3×10^{-3} of the touching static block at a lateral offset of $32h$, 5.6×10^{-3} at $64h$ and 1.1×10^{-2} at $128h$ ($\omega h/\beta_A = 1/6$). This is the far field of the waves along the interface, decaying as about h/r, with the P and S phases beating. It is not an artefact of the quadrature: the frame route agrees with the polar sum over the modes to 10^{-12} of the touching block at $8h$, $32h$ and $64h$. Sampling the spectrum of $\mathbf{D}$ on the wavenumbers of a periodic box would therefore add the images of this tail, an error of about h/P for a period P. Instead $\mathbf{D}$ is split by a C^∞ high-pass $\psi(q)$, which rises from twice the highest branch point (and at least $0.35/h$) to the start of the window, $Q_0/4$:

$$\mathbf{D} = \frac{1}{4\pi^2}\int (1-\psi)\,\mathbf{Y}\,\mathrm{d}^2k + \frac{1}{4\pi^2}\int_{q<Q_0} \psi\left[\mathbf{Y} - \chi(\mathbf{Y}_0 + \omega^2\mathbf{Y}_1)\right]\mathrm{d}^2k = \mathbf{D}_{\mathrm{low}} + \mathbf{D}_{\mathrm{high}}. \qquad (36)$$

$\mathbf{D}_{\mathrm{low}}$ holds the branch points and the tail. It remains the radial integral of Section 7 for each offset, but over $qh < 2.5$ only, with tables over that band. $\mathbf{D}_{\mathrm{high}}$ has a spectrum smooth in $\mathbf{k}$, so it decays faster than any power of the offset: below 10^{-8} of the touching block beyond 24 to 34 cells, and below 10^{-10} beyond 40 to 55 (the narrower band, $[1, 2.5]/h$ at $\omega h/\beta_A = 1/2$, decays more slowly than $[0.35, 2.5]/h$ at $1/6$). It is sampled at the wavenumbers $2\pi\mathbf{m}/(2Nh)$ of a periodic box of N cells and folded onto the box's offsets by Poisson's summation formula, on the wedge $0 \le k_y \le k_x$ alone: the eight signed permutations fixing z map the integrand as they map the blocks, which gives the rest of the grid exact to rounding. On $2 \times 4 \times 4$ cells, with sources on both sides, the split reproduces every block to 1.3×10^{-10} ($\omega h/\beta_A = 1/6$) and 2.9×10^{-9} ($1/2$) of the touching static block, and the GMRES solution to 7.4×10^{-12}. On 288 cells, measured in one process, the set-up takes 823 s (839 s unsplit, whose tables span $q < Q_0$), and each further frequency's kernels take 25 s against 67 s unsplit. The frequency-independent series coefficients of the whole-space blocks are computed with the set-up, so the first frequency costs about the same.

Without stored kernels. The transformed kernels of each pair of layers still grow as the square of the number of layers times $(2n-1)^2$. At a lateral wavenumber $\mathbf{k}$, however, the integrand of $\mathbf{D}$ separates in the two cells' depths. For a source in A it is the receiver's moments times the outgoing mode states, times the 3×3 matrix $\mathbf{R}_d$ or $\mathbf{T}_d$, times the three down-going modes' states and the source's moments. The window's static and ω^2 parts separate too, through the cells' vertical moments, with at most three orders on each side. Each factor is a cell's lateral moments times a mode map times its layer's phase $\mathrm{e}^{\mp \mathrm{i}k_z z}$, so only the layer-independent pieces are kept. A product then needs three amplitudes per wavenumber and source side, summed over the source layers and reconstructed at each receiver layer, and no block of a pair of layers is formed. The low band is applied at its polar quadrature nodes by non-uniform FFTs, which

keeps it exact and non-periodic; the high band at the wavenumbers of the periodic box, each alias its own node. In the same medium, cells in planes at least three apart couple through the modes of the whole space, factorised the same way about each cell's own centre and marched layer by layer with the step $\mathrm{e}^{\mathrm{i}k_z 2h}$, $|\cdot| \le 1$; between planes g apart the integrand falls as $\mathrm{e}^{-2(g-1)qh}$, so the march's wavenumbers stop at $qh = 8$. Only the whole-space blocks of planes 0, 1 and 2 apart stay stored, $5(2n-1)^2$ per side whatever the number of layers. Against Paper 2's blocks the march agrees to 3.4×10^{-10} ($\omega h/\beta_A = 1/6$) and 1.6×10^{-9} ($1/2$). The memory no longer grows with the number of layers: 2.8 and 2.9 GB for 24×24 cells in four and eight layers, where the stored kernels would take 1.8 and 7.3 GB.

Time is the price, and it is set by the number of wavenumbers. Three measures reduce it. First, the high band's nodes and the march's above the low band lie on the periodic box's lattice, $\mathbf{k} = 2\pi\mathbf{m}/(2Nh)$, where $\mathrm{e}^{-\mathrm{i}\mathbf{k}\cdot\mathbf{x}}$ over the cells is a discrete Fourier kernel: one zero-padded $N \times N$ FFT per layer gives every lattice node, aliases included, and the way back folds the nodes by $\mathbf{m}$ mod N and takes one inverse FFT. This is exact and needs no tolerance; it makes $\mathbf{D}_{\mathrm{high}}$ 4.3 times and the march 1.8 times faster for 24×24 cells. Only the polar nodes of the low band still need non-uniform FFTs. Second, their tolerance follows the error budget: at 10^{-9} a product stays within 10^{-11} of the stored kernels', and the transforms are 1.5 to 1.8 times faster than at 10^{-11}; larger transforms or a smaller upsampling factor gain nothing. Third, the low band's quadrature is tuned against the reference rule of Section 7: Gauss of order 8 instead of 12 on the plain radial panels and an azimuthal margin of 16 instead of 64 beyond the phase's bandwidth keep 33 to 56% of the nodes on 4×4 to 24×24 cells, the low band's product within 1.3×10^{-12}. The panels at the branch points keep order 24, which their square-root map needs, and the window's ramp its 16 panels, which narrow boxes need. With all three, products from the same set-up agree with those of the stored kernels to 1.7×10^{-11} on $2 \times 5 \times 5$ cells and on 96 cells in six layers (all whole-space couplings stored), and take 2.6 and 6.9 s (11.4 and 24.7 s before; peak memory 2.4 GB); on $8 \times 3 \times 3$ cells, where the march acts between planes three or more apart, to 1.5×10^{-11}, 8.2 s per product.

The iteration. Table 8 gives the products GMRES needs with the block preconditioner above, at relative tolerance 10^{-10}, for boxes straddling the interface with every cell at 10, 30 or 50% of its own medium's λ, μ and ρ, and the P wave above. At $\omega h/\beta_A = 1/6$ the count does not grow with the box; at $1/2$ and 50% it grows from 16 to 22 over sixteen times the lateral area, and more with depth than with width: four layers add more products than four times the area. Every case converged, and the matrix-free operator takes the same number of products as the stored kernels (16 on 72 cells at $1/2$ and 50%, the solutions 1.5×10^{-11} apart).

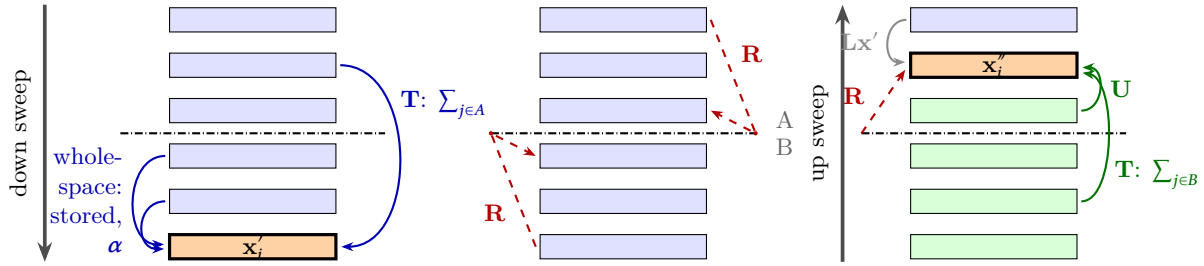
The two-way march as the preconditioner. Block-Jacobi ignores every coupling between planes, and its count grows with the number of layers. The plane-wave factorisation that applies the operator also gives a preconditioner that carries that coupling: the two-way march of the reflectivity method, read as a Gauss–Seidel iteration over the planes (Fig. 3). The coupling is split by the direction in which it travels, $\mathbf{K} = \mathbf{K}_0 + \mathbf{L} + \mathbf{U} + \mathbf{R}$: a plane with itself; from the planes above, the whole-space coupling of the side and the transmission from A into B;

Table 8. GMRES products (block-Jacobi, relative tolerance 10^{-10}) for boxes of layers $\times n \times n$ cells straddling the interface, at contrasts of 10/30/50%.

box	cells	$\omega h/\beta_A = 1/6$	$\omega h/\beta_A = 1/2$
$2 \times 6 \times 6$	72	8/10/12	9/13/16
$2 \times 12 \times 12$	288	8/11/13	10/15/19
$4 \times 12 \times 12$	576	8/11/14	11/17/24
$2 \times 24 \times 24$	1152	8/11/14	10/16/22

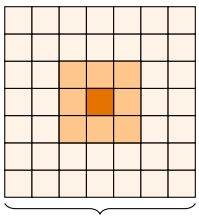
from the planes below, its mirror; and the reflections between different planes of one side. A down sweep solves the planes from the top, each with the field of the planes above already solved, $\mathbf{x}_i' = \mathbf{A}_i^{-1}(\mathbf{b}_i + \sum_{j<i} \mathbf{L}_{ij}\mathbf{E}\mathbf{x}_j')$ with $\mathbf{A}_i = \mathbf{M} - \mathbf{K}_{0,ii}\mathbf{E}$; the reflections of the result, $\mathbf{R}\mathbf{E}\mathbf{x}'$, are then evaluated once; and an up sweep solves the planes from the bottom with them, the down sweep's field from above and the up sweep's from below. Forward scattering is therefore exact within each sweep and the backscatter is lagged, as in the thesis's two-way march; further passes iterate the backscatter. The sweeps are the recursions already in the product: the march's amplitudes $\boldsymbol{\alpha}_i = p\,\boldsymbol{\alpha}_{i-1} + p^g\boldsymbol{\beta}_{i-g}$ carry the whole-space coupling down and up one plane at a time, the summed mode amplitudes of the side solved first carry the transmission, and each side's summed amplitudes carry its reflections, so a pass costs $O(L)$. Each solve is the intraplane step of Fig. 3(d): the plane's own system $\mathbf{A}_i$ couples only the cells of one plane, so it is a lateral problem. With the plane's own reflection lagged with the others (below), its product is the whole-space self kernel of the side, one lateral convolution applied by zero-padded two-dimensional FFTs, plus the corrections $\mathbf{A}_0 + \omega^2\mathbf{A}_1$ of the plane with itself, a local stencil; it is solved by an inner GMRES to 10^{-2}, preconditioned cell by cell with $(\mathbf{M} - \mathbf{K}_{mm}\mathbf{E}_m)^{-1}$, the right-hand side carrying the fields that the sweeps deliver to the plane. The inexact inner solve makes the preconditioner vary slightly between applications, so the outer iteration is flexible GMRES, which keeps the preconditioned vectors and needs no fixed preconditioner. Applied with the stored kernels of every pair of planes and through the recursions, the preconditioner agrees to 1.5×10^{-12} ($\omega h/\beta_A = 1/6$) and 1.9×10^{-11} (1/2) on $8 \times 2 \times 2$ cells, where the march acts.

Table 9 gives the outer products to 10^{-10} with four preconditioners, from the same stored kernels: none at all, the benchmark; block-Jacobi; the march, one pass; and Algorithm 5.1 for the plane-averaged contrast (below). Block-Jacobi gains less than it seems. Against no preconditioner it saves 5 to 9 products at $\omega h/\beta_A = 1/6$, but at 1/2 and 50% only 8 on two layers and 3 on sixteen (24 against 27); on $8 \times 12 \times 12$ cells 2 (27 against 29). The march takes a third to a half of the unpreconditioned count throughout. At 1/6 one pass makes it independent of the number of layers (4 to 8 products); at 1/2 it grows from 7 to 14 on two to sixteen layers, from the backscatter lagged once, and further passes remove the growth (6 products on two and on eight layers with three passes, on 6×6 cells). A preconditioner need not be exact, and its cost is set by the planes' own reflection and by the wavenumbers of $\mathbf{D}$. With each plane's own reflection lagged with the others, $\mathbf{A}_i$ is the plane's stored whole-space kernel and corrections, so the inner products are lateral FFTs; and the preconditioner's $\mathbf{D}$ and march get nodes of their own, the low band alone (the high band is local) at a coarse rule, non-uniform FFTs to 10^{-6} and the march's lattice to $qh = 4$. None of this changes the outer count, and a pass then costs about a quarter of



(a) planes from the top (b) reflections $\mathbf{R}\mathbf{E}\mathbf{x}'$, evaluated once (lagged) (c) planes from the bottom

(d) the plane's own system, $\mathbf{A}_i\mathbf{x}_i = \mathbf{M}\mathbf{x}_i - \mathbf{K}_{0,ii}\mathbf{E}\mathbf{x}_i$



whole plane (light): the whole-space self kernel of the side, one lateral convolution applied by zero-padded 2-D FFTs;
near cells (darker): the corrections $\mathbf{A}_0 + \omega^2\mathbf{A}_1$, a local stencil within the lateral radius;
the plane's own reflection: lagged with $\mathbf{R}$, so $\mathbf{A}_i$ needs no plane waves; solved by an inner GMRES to 10^{-2}, preconditioned cell by cell with $(\mathbf{M} - \mathbf{K}_{mm}\mathbf{E}_m)^{-1}$; the right-hand side is $\mathbf{b}_i$ plus the fields of (a)–(c).

Figure 3. One pass of the two-way march preconditioner, in a vertical section of planes of cells (A above the interface, dash-dotted; B below). (a) The down sweep solves the planes from the top; plane i receives the forward field of the planes already solved above it: the whole-space coupling of its side (stored kernels for touching planes, the march's amplitudes $\boldsymbol{\alpha}_i = p\,\boldsymbol{\alpha}_{i-1} + p^g\boldsymbol{\beta}_{i-g}$ beyond) and, in B, the transmission from all of A. (b) The reflections between different planes of one side, the backscatter, are evaluated once from the down sweep's solution. (c) The up sweep solves the planes from the bottom with the forward field from below, the transmission from B into A, the down sweep's field from above (stored) and the lagged reflections. (d) Each solve is the plane's own system, a lateral problem.

a product on 6×6 cells.

Algorithm 5.1 for the plane-averaged contrast. The march lags the backscatter; the thesis also gives the exact up/down sweep, its Algorithm 5.1, which factorises the depth problem, $\mathscr{A} = \mathscr{U}\mathscr{L}$, with Kennett's net reflection operators as pivots, so that a Riccati down-sweep forms the Schur complements $\mathbf{S}_z = \mathbf{D}_z - \mathbf{L}_z\mathbf{S}_{z-1}^{-1}\mathbf{U}_{z-1}$ and back-substitution goes up. $\mathbf{S}_z$ is the discrete Dirichlet-to-Neumann map of the half-domain above plane z: the backscatter of everything above, exactly. With a laterally varying contrast it is dense within each plane, but in a medium whose contrast varies in depth only it is diagonal in the lateral wavenumber. So each plane's contrast is replaced by its lateral mean, $\bar{\mathbf{E}}_z$: then $\mathbf{M} - \mathbf{K}\bar{\mathbf{E}}$ is invariant under lateral translations within each plane, separates by wavenumber on the stored kernels' padded grid (the circulant approximation of the box's convolutions), and at each wavenumber the depth system over the planes is solved exactly, here by a dense LU per bin, which is the same inverse. The backscatter of the averaged medium between planes is then exact, and only the lateral fluctuation of the contrast about its mean is left to the outer iteration. An application is an FFT per plane, a solve per wavenumber and an inverse FFT per plane, with no inner iteration. Its outer counts (Table 9) are those of the march to within one or two products. In those boxes every cell of a plane has the same contrast, so the plane's mean is its contrast, and what separates Algorithm 5.1 from an exact solve is the

Table 9. Outer products of FGMRES to 10^{-10} on the stored kernels, at contrasts of 10/30/50%, for boxes of $L \times n \times n$ cells: no preconditioner, block-Jacobi, the two-way march (one pass) and Algorithm 5.1 for the plane-averaged contrast.

$\omega h/\beta_A$	box	none	block-Jacobi	march	Alg. 5.1
1/6	$2 \times 6 \times 6$	14/18/21	8/10/12	4/6/6	5/6/6
	$4 \times 6 \times 6$	14/18/21	8/10/12	5/6/7	5/6/7
	$8 \times 6 \times 6$	14/18/21	8/11/13	5/6/7	5/7/8
	$16 \times 6 \times 6$	14/19/22	8/12/14	5/7/8	5/7/8
	$8 \times 12 \times 12$	14/19/22	9/12/15	5/7/8	5/7/9
1/2	$2 \times 6 \times 6$	15/20/24	9/13/16	5/7/7	5/7/8
	$4 \times 6 \times 6$	15/21/25	10/14/19	6/8/9	6/8/10
	$8 \times 6 \times 6$	15/21/26	10/16/21	6/9/12	6/9/13
	$16 \times 6 \times 6$	16/22/27	11/17/24	6/10/14	7/11/16
	$8 \times 12 \times 12$	16/23/29	11/19/27	6/10/14	6/10/15

box itself: the averaged medium fills the periodic grid, the cells only $n \times n$ of its $(2n-1)^2$ points. A homogenised layer, by the layer law, in place of each plane's mean contrast is the refinement that follows the thesis's own line of development, and is left for later.

Algorithm 5.1 in its Dirichlet-to-Neumann form. The exact factorisation of the depth problem, eliminated from the top, is $\mathscr{A} = \mathscr{L}\,\mathscr{S}\,\mathscr{U}$ with unit block-triangular factors and pivots $\mathbf{S}_z$, the Dirichlet-to-Neumann maps of the half-domains above. With the true contrast the pivots are dense within each plane; but a preconditioner may approximate them, and flexible GMRES accepts a preconditioner that is itself solved only partially. So the factors are taken from the plane-averaged medium, where at each wavenumber the block factorisation is exact and gives $\bar{\mathscr{L}}$, $\bar{\mathscr{U}}$ and the averaged pivots $\bar{\mathbf{S}}_z$, and each pivot takes the plane's own contrast back,

$$\tilde{\mathbf{S}}_z = \bar{\mathbf{S}}_z - \mathbf{K}_{zz}(\mathbf{E} - \bar{\mathbf{E}}_z), \tag{37}$$

with $\mathbf{E}$ the true contrast on the whole padded grid, $\mathbf{E}_m$ at the cells and zero at the points outside the box. The plane's contrast, and the box's edges, are then exact in its pivot, and only the backscatter from the planes above is the averaged medium's. An application is $\bar{\mathscr{U}}^{-1}\tilde{\mathscr{S}}^{-1}\bar{\mathscr{L}}^{-1}$: the triangular factors at each wavenumber, and each $\tilde{\mathbf{S}}_z^{-1}$ by an inner GMRES to 10^{-2}, preconditioned with $\bar{\mathbf{S}}_z^{-1}$. For a single plane, with a tight inner solve, it is the exact inverse. Table 10 compares the five preconditioners at $\omega h/\beta_A = 1/2$, with the contrast uniform in each plane and with a contrast varying from cell to cell, uniform on $[0, 50]$%. Where the contrast is uniform, the Dirichlet-to-Neumann form gains at most one product on the averaged one and is within one of the march. Where it varies, the averaged form degrades to 11 or 12 products, close to block-Jacobi's 12 to 17, the Dirichlet-to-Neumann form recovers most of the loss (8 to 10), and the march, which carries the true contrast everywhere, stays best (6 to 9). What separates the Dirichlet-to-Neumann form from the march is then the backscatter from above, still the averaged medium's.

The true pivot carries the planes above as well: $\mathbf{S}_z^{-1}$ is the block (z, z) of $(\mathscr{A}_{\le z})^{-1}$, the inverse

of the leading system of planes 1 to z. Solved exactly for every z it costs a full solve, so it is solved in a window, in the manner of sweeping preconditioners: the plane and the w planes above it with their true contrast, and the planes above the window through the averaged medium's Dirichlet-to-Neumann map, which is the trailing part of the averaged factorisation and costs nothing more. The pivot's inverse is then the plane's component of the window's system solved for a source on the plane, by an inner GMRES preconditioned with the window's averaged factorisation; $w = 0$ is the form above, and a window of all the planes above gives the exact pivots of the true medium, solved partially. Over all the planes the window's operator, the averaged trailing system less the contrast fluctuation, is the true system $\mathbf{M} - \mathbf{KE}$ on the padded grid; assembled independently from the stored kernels it agrees to 5.3×10^{-16}.

Table 10. Outer products of FGMRES to 10^{-10} on the stored kernels at $\omega h/\beta_A = 1/2$, for boxes of $L \times n \times n$ cells with every cell at 30% or 50% of its medium, or at a fraction uniform on $[0, 50]$% varying from cell to cell: no preconditioner, block-Jacobi, the two-way march (one pass), and Algorithm 5.1 for the plane-averaged contrast and in its Dirichlet-to-Neumann form (pivots to 10^{-2}).

box	contrast	none	block-Jacobi	march	Alg. 5.1, averaged	Alg. 5.1, DtN
$2 \times 6 \times 6$	30/50%	20/24	13/16	7/7	7/8	6/8
	varying	21	12	6	11	8
$4 \times 6 \times 6$	30/50%	21/25	14/19	8/9	8/10	8/10
	varying	22	14	7	11	9
$8 \times 6 \times 6$	30/50%	21/26	16/21	9/12	9/13	9/12
	varying	23	15	8	11	10
$16 \times 6 \times 6$	30/50%	22/27	17/24	10/14	11/16	10/15
	varying	24	16	9	12	10
$8 \times 12 \times 12$	30/50%	23/29	19/27	10/14	10/15	10/15
	varying	25	17	9	12	10

Table 11 gives the time to 10^{-10} with the matrix-free operator, where each outer product costs 12 to 41 s, for five boxes of three shapes. On the deep, narrow box ($32 \times 6 \times 6$, the shape of a teleseismic target) the march takes 27 and 25% less time than block-Jacobi with one pass and 37% less with three, and 42 to 48% less than no preconditioner; block-Jacobi there is barely better than none at 50% (27 products against 29). On boxes of moderate depth and width one pass saves 14 to 28%, and three passes beat one only at 50% on 6×6 cells per plane. On the wide, shallow box ($4 \times 24 \times 24$, the shape of a reflection survey) the march saves outer products as everywhere (10 and 14 against 19 and 28) but no time: one pass is level with block-Jacobi (406 against 387 s, 578 against 589 s) and three passes are slower. With four planes there is little coupling in depth for the march to carry and much within each plane, and a pass grows with the plane's width; the node bound below removes most of that growth but not the balance.

Why the nodes grow with the box, and why the preconditioner may bound them. Every application of $\mathbf{D}$, in the product and in the preconditioner, is a quadrature of its plane-wave integral, $\mathbf{D}(\boldsymbol{\rho}) \approx \sum_m w_m \, \mathbf{F}(\mathbf{k}_m) \, \mathrm{e}^{\mathrm{i}\mathbf{k}_m \cdot \boldsymbol{\rho}}$ over lateral wavenumbers $\mathbf{k}_m$, the nodes, and a pass of the march costs work at every node: the mode amplitudes, the non-uniform transforms and the contractions. The low band's nodes form a polar grid, and the rule must integrate the phase $\mathrm{e}^{\mathrm{i}\mathbf{k}\cdot\boldsymbol{\rho}}$

Table 11. Matrix-free FGMRES to 10^{-10} at $\omega h/\beta_A = 1/2$: outer products and time with no preconditioner, block-Jacobi and the two-way march (coarse nodes of its own, one or three passes), for boxes of $L \times n \times n$ cells at contrasts of 30% and 50%. One product takes 12 s ($8\times6\times6$), 25 s ($16\times6\times6$), 19 s ($8\times12\times12$), 41 s ($32\times6\times6$) and 20 s ($4\times24\times24$). †Outer products only, from the stored kernels (Table 9). ‡With the node bound (on 6×6 cells it changes only the march's lattice).

box	preconditioner	30%	50%
$8\times6\times6$	none	21†	26†
	block-Jacobi	16, 191 s	21, 247 s
	march, one pass	9, 138 s	12, 185 s
	march, three passes	8, 181 s	8, 176 s
$16\times6\times6$	none	22†	27†
	block-Jacobi	17, 404 s	24, 542 s
	march, one pass	10, 294 s	14, 414 s
	march, three passes	7, 303 s	8, 337 s
$8\times12\times12$	none	23†	29†
	block-Jacobi	19, 343 s	27, 486 s
	march, one pass	10, 295 s	14, 414 s
	march, three passes	7, 374 s	8, 433 s
$32\times6\times6$	none	23, 1026 s	29, 1237 s
	block-Jacobi	19, 847 s	27, 1133 s
	march, one pass	11, 615 s	15, 853 s
	march, three passes‡	7, 530 s	9, 717 s
$4\times24\times24$	none	25, 510 s	32, 647 s
	block-Jacobi	19, 387 s	28, 589 s
	march, one pass‡	10, 406 s	14, 578 s
	march, three passes‡	7, 617 s	9, 808 s

exactly for every lateral offset $\boldsymbol{\rho}$ in the box, up to the largest, the reach $\rho_{\max} \approx 2\sqrt{2}\,(n-1)h$ for $n \times n$ cells. Around a circle of radius q the phase oscillates about $q\rho_{\max}$ times, so the azimuthal trapezoid rule needs $n_\varphi \approx 2q\rho_{\max}$ points plus a margin; along q it oscillates with period $2\pi/\rho_{\max}$, so the radial panels narrow as the reach grows. The number of nodes therefore grows with the box, towards the square of the reach: from 14.9 to 24.6 thousand for the product's low band, and from 3.6 to 9.0 thousand for the preconditioner's coarse one, from 6×6 to 12×12 cells. The product cannot shed these nodes, since a rule too coarse for the reach aliases the field of a distant cell onto the wrong offsets and changes the solution. The preconditioner can. Its nodes may be held at the count for a fixed reach, whatever the box: its $\mathbf{D}$ is then accurate only between nearby cells, and in error for distant ones, where $\mathbf{D}$ itself is small (it decays as about h/ρ, to 5×10^{-3} of the touching block at an offset of $32h$). An inexact preconditioner can cost outer iterations but cannot change the answer, which the product fixes; so the bound trades the cost of a pass, then independent of the width, against the outer count. The coarsenings above left the count unchanged, which suggested the room to do so, and the measurement bears it out: on $4\times24\times24$ cells a bound at a reach of $32h$ (16 cells) cuts the preconditioner's low band from 28.2 to 9.3 thousand nodes with the outer count unchanged (10 and 14 products at 30 and 50%), and the solve's time by 10 to 13%.

Storage against time, over frequencies and sources. The matrix-free product exists to save memory, and it pays in time. A study over N_f frequencies with N_s right-hand sides each (incidence angles or sources) costs

$$T = S + N_f\big(B + N_s\, m\, t\big), \tag{38}$$

with S the frequency-independent set-up, B the operator's build for a frequency, m the outer products of a solve and t the time of one (plus the preconditioner's applications). Only S is amortised over the frequencies; B is paid at each frequency, and $m\,t$ for each source. The stored kernels and the matrix-free operator divide B and t oppositely (Table 12): stored kernels take longer to build, every pair of planes transformed, but a product costs hundredths of a second; the matrix-free operator builds in a fraction of the time, but a product costs seconds, a few hundred times more. With $m \approx 10$ the two are even at about one source per frequency, $N_s^* = (B_\mathrm{s} - B_\mathrm{mf})/(m\,(t_\mathrm{mf} - t_\mathrm{s})) \approx 1$ to 2 on these boxes; with more sources, as reflection studies have, the stored kernels win by a factor that grows with N_s. The matrix-free operator therefore does not make a frequency faster wherever the stored kernels fit; it is what makes a box fit at all. The stored kernels take $L^2(2n-1)^2$ blocks of 36×90 at degree one, $16L^2(2n-1)^2\cdot3240$ bytes: 0.4 GB for $8\times6\times6$ cells, 1.8 GB for $8\times12\times12$, 6.4 GB for $32\times6\times6$ and 54 GB for $64\times64\times8$. The matrix-free operator's memory does not grow with the number of layers; what it keeps is the whole-space kernels of touching planes, $5(2n-1)^2$ blocks per side, 8.4 GB for 64×64 cells, and the plane-wave factors. This is also why the preconditioner matters most here: each outer product it saves is worth seconds, not hundredths, at every frequency and for every source.

Table 12. The two operators from the same set-up at $\omega h/\beta_A = 1/6$: the set-up S (once), the build B for a frequency and the time t of one product, stored kernels against matrix-free.

box	S	B: stored / matrix-free	t: stored / matrix-free
$2 \times 5 \times 5$	243 s	42 s / 11 s	< 0.01 s / 2.6 s
$6 \times 4 \times 4$	412 s	94 s / 20 s	0.03 s / 6.9 s
$8 \times 3 \times 3$	247 s	167 s / 35 s	0.02 s / 8.2 s

8 Discussion

The static image carries the whole singularity of the coupling at a welded interface, and in closed form it costs no more than the touching blocks of a homogeneous medium: a fixed set of universal corner moments, contracted with exact rational weights and with coefficients rational in the Lamé constants. The dynamic remainder is smooth, and the plane waves of Section 2 carry it at a cost that does not depend on how close the cells are to the interface, since the subtraction has removed what made the integral converge slowly.

The split also preserves what makes a frequency sweep cheap in a homogeneous medium, where the closed-form blocks are set up once and each frequency costs little. Here the static blocks $\mathbf{K}_0^R$ and the coefficients $\mathbf{K}_1^R(Q_0)$ of the ω^2 term are computed once for each arrangement of cells, and at each frequency only the remainder is integrated, over $qh < Q_0 h$, with one interface solve per radial node and the angular integrals of the cells' lateral moments tabulated once, so that a frequency costs tens of milliseconds per block at $\omega h/\beta = 1/6$ and a few hundred at $\omega h/\beta = 1$. With the ω^4 term, Eq. (32), also in closed form for any media, the dropped remainder falls as Q_0^{-8}: at $Q_0 h = 10$ the worst of three pairs near the interface (the self-image pair, the pair touching across it and a lateral neighbour) is then exact to 4.8×10^{-11} at $\omega h/\beta = 1/6$ and 6.2×10^{-11} at $1/2$, against 3.5×10^{-10} and 2.5×10^{-8} with the ω^2 term alone. A lower cutoff would not make a frequency cheaper: the plane-wave part costs 0.03 to 0.06 s per pair at every $Q_0 h$ from 3 to 10, its work being the interface solve and the cells' moments at each radial node, not the length of the range. Nor does a solution see the gain: in the Galerkin solve of Section 7, at $\omega h/\beta = 1/6$, the scattered field of degree-one and degree-two cells differs by 2×10^{-4}, four or more orders above the error the ω^2 term leaves in the blocks, and toward $\omega h/\beta = 1$, where the gain grows, the cells are three per shear wavelength and the discretisation error grows faster. The ω^4 term is therefore a reference, for blocks exact to rounding at the cost of a closed-form $\mathbf{K}_2$ per arrangement of cells, not a default; higher orders follow the same way, each a polynomial two degrees higher, and are not needed for the same reason.

For a stack of layers the same construction applies interface by interface. The singular part of the reflected and transmitted couplings comes only from the interface that the cells touch, and is the static image of that interface; reflections from other interfaces reach a cell after travelling at least one layer thickness and are smooth. The plane-wave representation extends to the stack through the reflection and transmission matrices of the layered medium, composed by the recursions of the reflectivity method [2, 3]. The coupling of every cell with every other is then available from the up- and down-going factorisation alone, without tabulating the stratified Green's tensor, which is the storage advantage that motivates the approach. The

same factorisation gives the preconditioner: the two-way march in depth, forward scattering exact within each sweep and backscatter lagged, makes the iteration count nearly independent of the number of layers where block-Jacobi's grows, at a fraction of a product per pass.

One limitation remains: the closed form of the base integrals is stated for cubic corner boxes, $a = b = c$, which is all a grid of cubes requires; for unequal sides the angular integrals are evaluated by quadrature.

Data availability

The Mathematica notebooks and Python scripts that derive and check every result, and the package they test, are available at https://github.com/tmnestor/MultipleScatteringCalculations: the static image and transmitted field, `Mathematica/StaticInterfaceImage.wl`, `Mathematica/StaticImageSpace.wl` and `Mathematica/StaticInterfaceTransmission.wl`, with their Python twins `scripts/derive_static_interface_image.py`, `scripts/derive_static_image_space.py` and `scripts/derive_static_interface_transmission.py`; the canonical terms, `scripts/derive_interface_image_terms.py`; the corner moments and base integrals, `Mathematica/ImageCornerMoments.wl`, `scripts/derive_image_corner_moments.py` and `cubic_scattering/image_moments.py`; the blocks, `cubic_scattering/interface_image.py`; the plane-wave representation, reflection and transmission, `cubic_scattering/stratified_march.py`; the checks of Table 5, `scripts/gate_march_modes.py` and `scripts/gate_interface_image.py`; the term of order ω^2, `Mathematica/InterfaceOmega2.wl` and `scripts/derive_interface_omega2.py`; and the convergence studies of Tables 1 and 4, `scripts/measure_interface_static_subtraction.py` and `scripts/measure_interface_omega2_subtraction.py`; the split at many frequencies, `cubic_scattering/interface_spectral.py`, checked by `scripts/gate_interface_spectral.py`.

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This work began thirty years ago in the author's doctoral research at the Research School of Earth Sciences of the Australian National University, Canberra [4].

Declaration of generative AI and AI-assisted technologies in the manuscript preparation process

Software development. Software development followed a human-directed, AI-assisted workflow, organised as a Test Driven Hypothesis Acceptance cycle: each derivation, implementation or numerical claim was stated as a hypothesis together with a test designed to refute it (an independent route, an exact limit, a high-precision reference or a convergence study), and was accepted only when the test passed; a failed test returned the hypothesis for diagnosis, never the test for adjustment. The author designed the software originally, but only directed and

validated recent improvements, which build on the author's original Fortran implementation (the `PhD_fortran_code` folder of the repository given under Data availability) and an extensive library of Mathematica notebooks, developed and extended since 1992, that carry out and verify the symbolic derivations; an agentic coding assistant (Anthropic Claude) operated under human-in-the-loop supervision to implement specified changes and diagnostic tests. The design, the choice of numerical methods, and the resolution of the principal technical difficulties were the author's. All AI-generated changes were subject to code review by the author and validated by automated tests and independent checks.

Manuscript preparation. During the preparation of this work, the author used Claude (Anthropic) in order to search for relevant open-access research, to write the TikZ code of the diagrams and to help redraft the paper into more coherent sections for readability. After using this tool, the author reviewed and edited the content as needed and takes full responsibility for the content of the published article.

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